Future shifting with present marking in Khalkha Mongolian*

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Abstract

This paper examines a case of future temporal reference in Khalkha Mongolian (Eastern branch, Mongolic) that occurs in the absence of overt future morphology. When VPs are inflected solely with the tense morpheme *-nA*, either a present or future reading can arise, leading past descriptions of the language to characterize *-nA* as a non-past tense. However, the distribution of *-nA* raises questions for current semantic theories of the future: if a modal semantics is necessary for future temporal reference due to the uncertainty of future eventualities, does the tense morpheme give rise to the future-shifted readings on its own, or is a modal operator (covertly) present? I present novel empirical evidence that such a modal operator appears covertly alongside *-nA*, which I reclassify as a present tense due to its interactions with the Aktionsarten and aspect. While this modal is covert in affirmative sentences with *-nA*, I show that it is overtly realized under negation as the morpheme *-kh*. The availability of both present and future readings is traced to an ambiguity between two underlying structures: one that is future-oriented, and one that is present-oriented. In the future-oriented structure, future temporal reference stems from the presence of a covert modal operator and covert prospective aspect. Altogether, I argue that a modal analysis of future reference is needed, even when no modal appears on the surface.

Keywords: Future, Aspect, Modality, Present tense, Non-past tense, Khalkha Mongolian

1 Introduction

An unresolved issue in semantic theories of temporal reference concerns how future-oriented readings can arise in the absence of overt future marking (for an overview, see Bochnak, 2019).

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This paper focuses on such a case of future temporal reference in Khalkha Mongolian (Eastern branch, Mongolic), where sentences without future marking can be interpreted as future-shifted. In Khalkha Mongolian, when the tense suffix *-nA* attaches to a verbal root like in (1), either a present or future interpretation can arise.[1] The temporal interpretation of sentences containing *-nA* is dependent on the Aktionsart of the VP that *-nA* attaches to: with eventive VPs (1a), only a future reading is possible, but with stative VPs (1b), either a present or future reading is available.

(1) a. Tselmeg shüleg unsh-**ina**.
Tselmeg poem read-*nA*
'Tselmeg *reads/will read a poem.'

b. Bat minii nuuts-iig med-**ne**.
Bat 1SG.POSS secret-ACC know-*nA*
'Bat knows/will know my secret.'

The availability of both present and future readings has led past descriptions of Khalkha Mongolian to characterize *-nA* as a non-past tense marker (Binnick, 2012, 2016; Song, 1997; Svantesson, 1991), wherein the non-past tense introduces a reference time that is equivalent to or follows the time of speech. Other languages have also been characterized as having past versus non-past tense distinctions, such as German (Hilpert, 2008; Musan, 2002; Klein, 1994) and Japanese (Kaufmann & Miyachi, 2011; Aonuki, 2021).

However, the existence of a non-past tense category raises questions for the semantics of future reference more generally. While some analyses opt for a non-modal treatment of futures like English *will* (e.g., Kissine, 2008), others argue that the future is inherently uncertain, thus requiring the presence of a modal operator rather than a purely temporal operator (e.g., Thomason, 1970; Condoravdi, 2002; Bochnak, 2019). If non-past tenses contribute forward temporal shifting on their own, do they co-occur with a covert modal operator, or are they a case of non-modal future reference?

Further, the presence or absence of modality in sentences like (1) has consequences for determining the semantics of *-nA* itself. In the absence of a future-oriented modal, a non-past tense analysis of *-nA* is necessary to achieve the future-oriented readings of (1). This is because a present tense is unable to contribute a future reference time on its own, unlike the non-past. Alternatively, if a future-oriented modal operator were covertly present in (1), it would remove any need to treat the tense category of *-nA* as non-past rather than present. Instead, the necessary future shifting would stem from this modal element, giving rise to the future readings in (1).

To address this question, I investigate the distribution of future readings with *-nA* in Khalkha Mongolian. While both present and future temporal reference are available with *-nA*, I argue that its distribution is incompatible with a non-past tense analysis and provide evidence that it is, instead, a present tense marker. How, then, do future readings arise if the tense is present, rather than non-past? I discuss data on the morphosyntax of negation and tense which supports the presence of a covert modal element in such cases: though this modal is covert in affirmative sentences, I argue that it is overtly realized under sentential negation as the morpheme *-kh*. In addition, because the modal *-kh* does not enforce future shifting, I propose that a covert

[1] When represented orthographically in Khalkha Mongolian, this tense suffix is written with a harmonized vowel following the nasal consonant. However, this vowel is typically unpronounced.

prospective aspect appears alongside it, following recent decompositional accounts of future reference (Bochnak, 2019; Kratzer, 2011; Mucha, 2016; Rivero, 2023).

The proposed structure for sentences with *-nA* that receive future readings is shown in (2a). It is ambiguous with another structure in (2b) that results in present-orientation but is infelicitous with eventive VPs, which I trace to the 'present perfective paradox' (Bennett & Partee, 1972; de Wit, 2016; Malchukov, 2009; Ogihara, 2007).

(2) a. Future-oriented structure:
$[_{TP}\ [_{ModP}\ [_{AspP}\ [_{AspP}\ [_{vP}\ \ldots]\ _{Asp}\ \textsc{prfv}]\ _{Asp}\ \textsc{prosp}]\ _{Mod}\ \textsc{mod}]\ _{T}\ \textsc{prs}]$
b. Present-oriented structure:
$[_{TP}\ [_{AspP}\ [_{vP}\ \ldots]\ _{Asp}\ \textsc{prfv}]\ _{T}\ \textsc{prs}]$

The structure of the paper is as follows: in Section 2, I outline my preliminary theoretical assumptions on the semantics of tense, aspect, and modality and give relevant background on Khalkha Mongolian. In Section 3, I provide data on the distribution of present and future readings with *-nA* in Khalkha Mongolian, with particular attention given to the role of the Aktionsarten and aspect. Based on its distribution, I argue for an analysis of *-nA* as a present tense. In Section 4, I present a formal account of future reference in Khalkha Mongolian, where sentences with *-nA* are ambiguous between the two structures in (2). The structure in (2a) results in future shifting because the present tense appears above a modal and prospective aspect, both of which are covert in affirmative sentences. In Section 5, I give explicit evidence that future readings of sentences with *-nA* are possible via the presence of this covert modal, which overtly surfaces under negation. In Section 6, I extend the analysis of future shifting with *-nA* in affirmative sentences to their negative counterparts. Section 7 concludes by discussing the implications of this analysis for the status of non-past tenses.

2 Preliminaries

This section lays out the preliminary theoretical assumptions taken in this paper (Section 2.1), as well as relevant background on Khalkha Mongolian (Section 2.2).

2.1 Theoretical preliminaries

I presuppose a neo-Reichenbachian framework for tense and aspect (Klein, 1994; Reichenbach, 1947). Under this framework, tense determines the temporal relation between the *utterance time* (UT) and *reference time* (RT).

For the purposes of illustration, I use a quantificational semantics for tense in this paper: tense introduces existential quantification over times, the domain of which is contextually restricted (see Ogihara & Kusumoto, 2020). An example is given for the past tense in (3). In (3), *g(i)* is a contextually-salient interval, and its superset relation with *t'*, the RT, restricts the domain of quantification. The proposal outlined in this paper is also compatible with a pronominal theory of tense (Kratzer, 1998; Partee, 1973, 1984).

(3) $[\![\textsc{pst}_i]\!]^{g,c} = \lambda p.\exists t'[\ t' < t_c\ \&\ t' \subseteq g(i)\ \&\ p(t')\ ]$

A quantificational semantics for the present tense is shown in (4). The present in (4) differs from the past in (3) by virtue of *t'* being equivalent to, rather than preceding, t_c.

(4) $[\![\text{PRS}]\!]^{g,c} = \lambda p.\exists t'[\ t' = t_c\ \&\ p(t')\ ]$

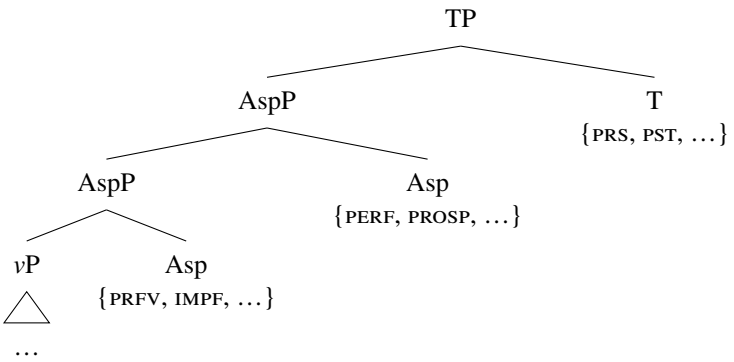
Viewpoint aspect is concerned with temporal relations between an RT and *eventuality time* (ET), demonstrated with lexical entries for the perfective in (5a) and the imperfective in (5a). Structurally, I take viewpoint aspect to appear at a position above vP. For VPs, I assume a classification of the Aktionsarten where VPs may be interpreted as states, activities, accomplishments, or achievements. I use the term *eventive* to refer to the latter three classes as a whole.

(5) a. $[\![\text{PRFV}]\!]^{g,c} = \lambda P.\lambda t.\lambda w.\exists e[\ \tau(e) \subseteq t\ \&\ P(e)(w)\ ]$
b. $[\![\text{IMPF}]\!]^{g,c} = \lambda P.\lambda t.\lambda w.\exists e[\ t \subset \tau(e)\ \&\ P(e)(w)\ ]$

I adopt a two-layered approach to aspect where, in addition to viewpoint aspect, *high aspect* exists as a separate type of aspect. The syntactic position of high aspect is above viewpoint aspect, but below tense. High aspect relates the RT introduced by tense to another RT, and, therefore, does not directly encode any temporal relation with the ET. Proposed types of high aspect include the perfect (Iatridou et al., 2001; Pancheva, 2003), as well as the prospective (Pancheva & Zubizarreta, 2023). I discuss the semantic consequences of high versus viewpoint aspects further in Section 4.1.

I summarize my assumptions for the structure of the verbal domain in (6), represented head-finally to reflect the syntax of Khalkha Mongolian.

(6)



As a final preliminary assumption, I adopt a Kratzerian treatment of modality (Kratzer, 1981). As such, modals operate over possible worlds, constrained by a modal base and ordering source. I assume both the modal base and ordering source receive their specification via a contextually-supplied conversational background.

2.2 Language background and methodology

Khalkha Mongolian, like other Mongolic languages, is a head-final language with agglutinative morphology. Most relevantly, Khalkha Mongolian possesses a rich morphological system for the expression of tense and aspect. As background, I briefly summarize its inventory of tense/aspect suffixes.

In Khalkha Mongolian, tense(-like) morphemes can either be non-evidential, or contribute an evidential meaning. I concentrate on the non-evidential tenses in this paper. Non-evidential tense morphology includes the suffix *-nA*, whose semantics is the focus of the current paper, and the past tense *-sAŋ*. Examples of evidential 'tenses,' on the other hand, are the direct proximate *-lAA* and the indirect past *-jee* (e.g., Binnick, 2012; Song, 1997).

In addition to its tense(-like) morphology, the language has several aspectual suffixes. These suffixes appear in a syntactically lower position than tense and form verbal participles. The aspectual suffixes consist of the habitual *-dAg*, the imperfective *-(ɢ)AA*, the past participle *-sAŋ*, and the "future" participle *-kh* (e.g., Binnick, 2012; Brosig, 2015). I point out that like English *-ed*, the form *-sAŋ* in Khalkha Mongolian leads a double life as both a finite past tense morpheme and a past participial suffix. The morphosyntax of these suffixes, as well as the semantics of the suffix *-kh*, will be described in greater depth in Section 5.

The uncited Khalkha Mongolian data were collected in consultation with four native speakers of the language, using standard methods for semantic elicitation. Tasks used to elicit semantic data included discussing the acceptability of sentences, judgments on their truth and felicity within discourse contexts, and translation between English and Khalkha Mongolian as needed (see Matthewson, 2004; Cover, 2015). Elicitations were conducted both in-person in California and remotely via video call. Each judgment was verified by two of the Khalkha Mongolian speakers following initial data collection.

3 Future reference with *-nA* in Khalkha Mongolian

I begin this section by describing the distribution of *-nA* and the conditions that govern its temporal reference (Section 3.1). Based on these observations, I argue that *-nA* is a present tense, rather than a non-past (Section 3.2).

3.1 The distribution of *-nA*

In Section 3.1.1, I demonstrate that the interpretation of *-nA* interacts with the Aktionsart of the VP it attaches to. More precisely, the temporal interpretation of the sentence depends on whether the verb is eventive (future-oriented only) or stative (future- or present-oriented). I then show that alongside the Aktionsart of the VP, aspect also determines what temporal reference is available with *-nA*: a future interpretation is only available with the perfective aspect (Section 3.1.2). In contrast, for the perfect or the imperfective, the sentence must be interpreted as present (Section 3.1.3). I show that modal subordination is licensed for sentences with *-nA* (Section 3.1.4). Finally, I compare the distribution of *-nA* to other cross-linguistic phenomena involving future reference with present morphology (Section 3.1.5).

3.1.1 The Aktionsarten and *-nA*

When attached to eventive VPs like those in (7), *-nA* yields only future readings. Such sentences cannot be interpreted as present-oriented, as shown by their incompatibility with an indexical interpretation of present-oriented temporal adverbials like *odoo* 'now' in (7a) and

(7c).[2] In comparison, future-oriented temporal adverbials are felicitous, such as *margaash* 'tomorrow' in (7b) and (7d).

(7) a. #Tselmeg odoo shüleg unsh-**ina**.
Tselmeg now poem read-*nA*
Intended: 'Tselmeg reads a poem now.'

b. Tselmeg (margaash) shüleg unsh-**ina**.
Tselmeg (tomorrow) poem read-*nA*
'Tselmeg will read a poem (tomorrow).'

c. #Suvdmaa odoo büjigle-**ne**.
Suvdmaa now dance-*nA*
Intended: 'Suvdmaa dances now.'

d. Suvdmaa (margaash) büjigle-**ne**.
Suvdmaa (tomorrow) dance-*nA*
'Suvdmaa will dance (tomorrow).'

Unlike eventive VPs, stative VPs that appear with *-nA* can be interpreted as either present- or future-oriented, as in (8). Statives marked with *-nA* are compatible with both present-oriented (8a, 8c) and future-oriented temporal adverbials (8b, 8d).

(8) a. Bat odoo minii nuuts-iig med-**ne**.
Bat now 1SG.POSS secret-ACC know-*nA*
'Bat knows my secret now.'

b. Bat margaash minii nuuts-iig med-**ne**.
Bat tomorrow 1SG.POSS secret-ACC know-*nA*
'Bat will know my secret tomorrow.'

c. Khöshöö odoo uul-in oroi-d bairla-**na**.
statue now mountain-GEN top-DAT be.placed-*nA*
'The statue is placed at the top of the mountain now.'

d. Khöshöö margaash uul-in oroi-d bairla-**na**.
statue tomorrow mountain-GEN top-DAT be.placed-*nA*
'The statue will be placed at the top of the mountain tomorrow.'

3.1.2 The perfective and *-nA*

In addition to the Aktionsart of the VP, the temporal reference of *-nA* is also sensitive to aspect. While the aforementioned future-shifted sentences with *-nA* lack overt aspectual morphology, such sentences are interpreted as having a perfective viewpoint aspect.

To demonstrate this, consider sentences with *when*-clauses like (9), where the *when*-clause explicitly provides a punctual RT (e.g., Smith, 1991; Wurmbrand, 2014). In (9a), Khulan's watering of the plants cannot be interpreted as simultaneous with Tuya's arrival. The only available reading is one where these events are sequential, which is characteristic of the perfective. In (9b), where the verb is marked as imperfective, the events are instead interpreted as simultaneous: Khulan's watering of the plants overlaps with Tuya's arrival. For future shifting

[2](7a, 7c) are felicitous under a non-indexical interpretation of 'now' (e.g., Altshuler, 2016), where only a future-oriented reading is allowed (i.e., "Tselmeg will read a poem now").

with imperfectives, the verb is marked with the converb *-j* and co-occurs with a participial form of the auxiliary *baikh* 'to be'. A higher verb, *bolokh* 'to become', is then inflected with *-nA*.

(9) a. Tuya tsetserleg-t khüreelen-d ochi-kh üye-d, Khulan urgamal usal-**na**.
Tuya garden-LOC park-DAT arrive-*kh* time-DAT Khulan plant water-*nA*
'When Tuya arrives at the garden, Khulan will water the plants.' (Sequenced only)
≠ 'When Tuya arrives at the garden, Khulan will be watering the plants.'

b. Tuya tsetserleg-t khüreelen-d ochi-kh üye-d, Khulan urgamal usal-j
Tuya garden-LOC park-DAT arrive-*kh* time-DAT Khulan plant water-CVB
bai-kh bol-**no**.
AUX-MOD become-*nA*
'When Tuya arrives at the garden, Khulan will be watering the plants.

The contrast between (9a) and (9b) arises because the *when*-clause introduces an RT for the main clause that is shortly after the eventuality the *when*-clause describes (Hinrichs, 1986; Partee, 1984). Specifically, in (9), the RT of the main clause, t_{main}, closely follows the RT of the *when*-clause, t_{when} (i.e., the time of Tuya's arrival). This property of *when*-clauses interacts with the semantics of the perfective and imperfective as follows: the semantics of the perfective states that the eventuality's duration is contained within the RT of the clause it appears in. Thus, for (9a), the duration of Khulan's watering of the plants is contained within t_{main}. Because t_{main} follows t_{when} and does not overlap with it, this means the eventualities contained by t_{main} and t_{when} do not overlap in time themselves. As a result, if the main clause has the perfective aspect, a sequenced reading is expected where Khulan's watering of the plants follows Tuya's arrival. This is the reading observed for (9a).

The temporal relation encoded by the semantics of the imperfective, in contrast, is reversed: it states that the ET contains the RT of the clause. As a result, in (9b), t_{main} is contained within the duration of Khulan's watering of the plants. By virtue of the imperfective's containment relation, this eventuality must also hold, minimally, at the times that immediately flank t_{main}. Since t_{main} is directly preceded by t_{when}, it overlaps with Khulan's watering of the plants. Thus, Khulan's watering of the plants is interpreted as occurring at the same time as Tuya's arrival in (9b).

The same behavior is observed for sentences marked with the past tense *-sAŋ*, as illustrated by (10). With no other verbal morphology present, a verb marked with *-sAŋ* in the main clause can only lead to a sequenced reading. Complementing the data on *-nA* and *-sAŋ* given here, Munkhbat (2024) identifies a similar instance of perfectivity in Khalkha Mongolian for verbs that are marked with the direct evidential *-lAA* and no additional aspectual morphology. This evidence thus supports the proposal that sentences which are aspectually bare on the surface are interpreted as perfective in Khalkha Mongolian, regardless of what tense they are marked with. This parallels the behavior of 'simple present/past' sentences in English, where similar findings have been used to argue for the presence of a covert perfective (Arregui, 2007; Wurmbrand, 2014).

(10) a. Tuya tsetsleg-t khüreelen-d ochi-kh üye-d, Khulan urgamal usal-**saŋ**.
Tuya garden-LOC park-DAT arrive-*kh* time-DAT Khulan plant water-PST
'When Tuya arrived at the garden, Khulan watered the plants.' (Sequenced only)

≠ 'When Tuya arrived at the garden, Khulan was watering the plants.'

b. Tuya tsetsleg-t khüreelen-d ochi-kh üye-d, Khulan urgamal usal-j
Tuya garden-LOC park-DAT arrive-*kh* time-DAT Khulan plant water-CVB
bai-**saŋ**.
AUX-PST
'When Tuya arrived at the garden, Khulan was watering the plants.'

Further support for the perfectivity of sentences where the verb is marked with *-nA* comes from their infelicity when a punctual RT has been specified in a preceding question context (Mucha, 2016). I first demonstrate this for future marking via WOLL in English using examples from Mucha (2016), shown in (11).

(11) Speaker A: Can I meet Alex tomorrow at 6:00pm sharp?

a. No, Alex will be working.

b. #No, Alex will work. (from Mucha, 2016)

In (11), the adverbial in the question sets the RT to 6:00pm. WOLL is unacceptable in response to such questions when it appears above a bare verb (11b). The only acceptable response with WOLL is one where the lower verb is inflected with *-ing* (11a), indicating imperfectivity.

The way this pattern is borne out stems again from the different temporal relations encoded by the perfective versus imperfective aspects. As discussed before, the perfective expresses a relation where the RT contains the eventuality's duration. However, in (11b), the RT is a single point in time. Because the duration of the eventuality (i.e., Alex's working) is not instantaneous and thus cannot be contained by a single time point, the perfective sentence in (11b) is infelicitous. The imperfective sentence in (11a), on the other hand, is felicitous, since the punctual RT can be contained by the ET.

Khalkha Mongolian sentences with *-nA* in a similar question context are given in (12). Only the imperfective aspect in (12a) is felicitous. (12b), where *-nA* appears without additional aspectual morphology, is infelicitous. The unacceptability of (12b) indicates that, like the English WOLL example in (11a), its aspect is perfective, resulting in infelicity.

(12) A: Can I visit Oyuun tomorrow at 6:00pm sharp?

a. B: Ügüi, Oyuun Ulaanbaatar luu ayal-aj bai-kh bol-**no**.
No, Oyuun Ulaanbaatar towards travel-CVB AUX-*kh* become-*nA*
'No, Oyuun will be traveling to Ulaanbaatar.'

b. B: #Ügüi, Oyuun Ulaanbaatar ruu ayal-**na**.
No, Oyuun Ulaanbaatar towards travel-*nA*
Intended: 'No, Oyuun will travel to Ulaanbaatar.'

Based on the data in (9–12), I conclude that when the verb is aspectually bare on the surface, it is interpreted as perfective. Under this view, the future-shifted examples with *-nA* like (1–8) possess a covert perfective morpheme. For eventive VPs, only future readings of *-nA* are available with the perfective aspect. In contrast, for their stative counterparts, both present and future readings are possible. I return to the status of stative VPs in Section 4.2.2, where I propose that in Khalkha Mongolian, statives are able to appear with the perfective.

3.1.3 The interpretation of *-nA* with other aspects

Unlike the perfective sentences, *-nA* can only result in present temporal reference for other aspects, such as the perfect and the imperfective. In Khalkha Mongolian, the perfect is formed via the auxiliary *baikh* 'to be', which appears in a position above the past participle *-sAŋ*. To express the present perfect, *-nA* appears on this higher auxiliary, as shown in (13). In these constructions, regardless of whether the verb is eventive (13a) or stative (13b), only present readings are possible. Future-oriented adverbials are disallowed in both cases.

(13) a. Tselmeg (#margaash) shüleg unsh-saŋ bai-**na**.
Tselmeg (tomorrow) poem read-PST.PRTCP AUX-*nA*
'Tselmeg has read a poem (#tomorrow).'
≠ 'Tselmeg will have read a poem tomorrow.'

b. Bat (#margaash) minii nuuts-iig med-seŋ bai-**na**.
Bat (tomorrow) 1SG.POSS secret-ACC know-PST.PRTCP AUX-*nA*
'Bat has known my secret (#tomorrow).'
≠ 'Bat will have known my secret tomorrow.'

The temporal reference of *-nA* is also unambiguous with the imperfective aspect, like in (14). The imperfective is realized via the converb *-j*, which is attached to the verb, and the auxiliary *baikh* 'to be', which appears higher in the structure and inflects for tense.[3] As with the perfect examples in (13), both eventive (14a) and stative (14b) verbs are interpreted as present, and future-shifted readings are unavailable. This is exemplified once again by the infelicity of future-oriented adverbials.[4]

(14) a. Tselmeg (#margaash) shüleg unsh-ij bai-**na**.
Tselmeg (tomorrow) poem read-CVB AUX-*nA*
'Tselmeg is reading a poem (#tomorrow).'
≠ 'Tselmeg will be reading a poem (tomorrow).'

b. Bat (#margaash) minii nuuts-iig med-ej bai-**na**.
Bat (tomorrow) 1SG.POSS secret-ACC med-CVB AUX-*nA*
Literal: 'Bat is knowing my secret (#tomorrow).'
≠ 'Bat will be knowing my secret (tomorrow).'

3.1.4 Modal subordination with *-nA*

Modal subordination is licensed for sentences with *-nA*, suggesting that futurity with *-nA* cannot be achieved through purely temporal means. Modal subordination refers to the phenomenon where the prejacent of a modal operator is semantically subordinated by a modal operator in a previous sentence, leading to restrictions on presupposition satisfaction (Roberts, 1989). English examples are shown below in (15), adapted from Roberts (1989).

[3] In addition to the converbial imperfective construction in (14), descriptions of Khalkha Mongolian also report a dedicated imperfective morpheme *-(ɢ)AA*. In affirmative sentences, *-(ɢ)AA* is typically only acceptable with the auxiliary *baikh* and rarely, if ever, can it attach to other matrix verbs (e.g., Binnick, 2012; Song, 1997).

[4] The acceptability of stative VPs here indicates that the examples in (14) are not progressive like English *-ing*, which often leads to infelicity with stative predicates (Lakoff, 1966; Dowty, 1975), but are instead better treated as imperfective (e.g., Deo, 2009).

(15) a. If John bought a book, he will be reading it at home later. It will be a murder mystery.

b. If John bought a book, he was reading it at home yesterday. #It was a murder mystery.

In (15), the DP *a book* can only serve as an antecedent for the pronoun *it* in the second sentence if a modal operator is present, like WOLL in (15a). Otherwise, without a modal, such anaphora is infelicitous. By virtue of its ability to diagnose modal elements, modal subordination has been used to argue for the modal status of future expressions (Klecha, 2014).

The same contrast holds between sentences with *-nA* in (16a) and sentences with a past tense and no modal operator in (16b). The former is a felicitous continuation, whereas the latter is only acceptable if an epistemic modal is added, realized via the suffix *-kh* appearing on a higher auxiliary. Modal subordination examples like (16a) pose difficulties for an analysis that does not assume the presence of a modal operator alongside *-nA*: if *-nA* is indeed non-modal, why does modal subordination occur for sentences with *-nA* where no modal operators are present on the surface?

(16) a. Context: Your friend Enkhbayar is performing at a concert later. You don't know what instrument he'll be playing, but Enkhbayar is only good at playing specific instruments.

Enkhbayar gitar toglo-vol, kontsert saikhan bol-no. Üzegchid ilüü alga
Enkhbayar guitar play-COND concert nice become-*nA* audience more palm
tash-**ina**.
clap-*nA*
'If Enkhbayar plays the guitar, the concert will be nice. The audience will applaud more.'

b. Context: Your friend Enkhbayar performed at a concert yesterday. You didn't go to the concert, so you don't know what instrument he played, but Enkhbayar is only good at playing specific instruments.

Enkhbayar gitar toglo-soŋ bol, kontsert saikhan bol-soŋ bai-kh. Üzegchid
Enkhbayar guitar play-PST COND concert nice become-PST AUX-*kh* audience
ilüü alga tash-**saŋ #(bai-kh)**.
more palm clap-PST AUX-*kh*
'If Enkhbayar played the guitar, the concert must have been nice. The audience #(must have) applauded more.'

3.1.5 A comparison with other future readings of the present

Before turning to the proposal for the semantic contribution of *-nA*, I compare the conditions that license future readings with *-nA* to other attested future readings of present tense sentences. In several languages, future readings can obtain for sentences with present morphology. In particular, futurates (Lakoff, 1971; Vetter, 1973; Huddleston, 1977; Copley, 2002, 2008; Rullmann et al., 2023) and future readings in subordinate clauses (Kaufmann, 2005; Mendes, 2024; Schulz, 2008; Crouch, 1994; Rumberg & Lauer, 2023) both result in future-orientation despite having present tense forms. Examples are given in (17a) and (17b), respectively.

(17) a. The San Francisco Orchestra performs tomorrow night.
b. If the orchestra plays the symphony with no mistakes, they practiced diligently.

Given this, a sensible hypothesis may be that the future readings with *-nA* are a similar species to futurates or subordinate futures. However, the availability of both is restricted: futurates require that the eventuality is planned or scheduled to occur (Copley, 2002, 2008), and the future that arises in subordinate clauses is, in fact, limited to such subordinate clauses (Mendes, 2024). These restrictions do not apply to future readings with *-nA* in Khalkha Mongolian. In the absence of a planning context, the English futurate in (18a) is infelicitous, whereas the Khalkha Mongolian sentence in (18b) is felicitous.

(18) a. #It snows tomorrow.
b. Margaash tsas or-**no**.
tomorrow snow enter-*nA*
'It will snow tomorrow.'

While subordinate future readings are felicitous only in subordinate clauses, like the antecedent of a conditional in (19a), such future interpretations are infelicitous in matrix clauses like (19b). In contrast, for verbs marked with *-nA*, future interpretations like in (19c) are freely available in matrix clauses.

(19) Context: You're talking to your friend about an upcoming concert, where your other friend Enkhbayar is performing.
a. If Enkhbayar plays the saxophone, forgetting my ear plugs was a mistake.
b. #Enkhbayar plays the saxophone. (Present-oriented habitual only)
c. Enkhbayar saksofon toglo-**no**.
Enkhbayar saxophone play-*nA*
'Enkhbayar will play the saxophone.'

Thus, the Khalkha Mongolian case constitutes a separate type of future reading, distinct from futurates and subordinate futures.

3.2 *-nA* as a present tense marker

While the availability of both present and future reference with *-nA* has led prior Khalkha Mongolian descriptions to label it as a non-past tense marker (e.g., Binnick, 2012; Song, 1997; Svantesson, 1991), I argue that this characterization is unable to capture its behavior. In Section 3.2.1, I show that a non-past tense account makes incorrect predictions regarding the interpretation of *-nA*. I then propose that *-nA* should instead be treated as a present tense (Section 3.2.2).

3.2.1 Against a non-past analysis of *-nA*

To evaluate the predictions of a non-past tense account, NON-PAST is formalized in (20). Under (20), a time *t'* is introduced that serves as the RT, and no subpart of *t'* may precede the UT. In other words, *t'* is either equivalent to or follows the UT. This treatment of NON-PAST reverses the NON-FUTURE tense semantics given by Matthewson (2006) for St'át'imcets.

(20) a. $t' \geq t := \neg\exists t''[\ t'' \subset t' \ \&\ t'' < t\ ]$

b. $[\![\text{NON-PAST}]\!]^{g,c} = \lambda p.\exists t'[\ t' \geq t_c \ \&\ p(t')(w_c)\ ]$

Though (20) correctly predicts the availability of both present and future reference with *-nA*, there are two patterns that it cannot predict: the interaction between temporal orientation and aspect (see Section 3.1.2), and the inability of *-nA* to introduce an RT that covers both the present and future (cf. Matthewson, 2006).

To start, consider how the semantics in (20) would interact with aspect. If *-nA* contributed a non-past RT like in (20), future-shifted readings would be predicted to arise regardless of aspect, rather than be restricted to the perfective. As an example, let's walk through the predicted truth conditions for NON-PAST with the perfect aspect. To do so, I first discuss the meaning of the perfect that will be assumed for the sake of illustration. In (21), a 'weak' extended now (XN) semantics for the perfect is provided, following Pancheva & von Stechow (2004).[5]

Under an XN theory of the perfect (e.g., McCoard, 1978; Dowty, 1979; Iatridou et al., 2001), the perfect introduces a time interval known as the perfect time span (PTS). The relation encoded by $\leq$, defined in (21a), requires that no subpart of the PTS follows the RT contributed by tense.[6] The lexical entry for PERF is given in (21b).

(21) a. $t' \leq t := \neg\exists t''[\ t'' \subset t' \ \&\ t'' > t\ ]$

b. $[\![\text{PERF}]\!]^{g,c} = \lambda p.\lambda t.\exists t'[\ t' \leq t \ \&\ p(t')\ ]$

The perfect, taken to be a high aspect, scopes above viewpoint aspect and relates the RT introduced by tense to another RT (Iatridou et al., 2001; Pancheva, 2003; Pancheva & Zubizarreta, 2023). Therefore, when viewpoint aspect appears under the perfect, it expresses a relation between the PTS (i.e., the RT introduced by the perfect) and the ET. Different readings of the perfect (e.g., universal, experiential, etc.) stem from the particular viewpoint aspect that appears below it (Iatridou et al., 2001; Pancheva, 2003). A sentence like (13a), repeated below in (22), receives an experiential perfect reading: it is entailed that the eventuality holds at a subset of the PTS.

(22) Tselmeg shüleg unsh-saŋ bai-**na**.
Tselmeg poem read-PST.PRTCP AUX-*nA*
'Tselmeg has read a poem.'
$\neq$ 'Tselmeg has been reading a poem.'

The experiential reading of the perfect is modeled in (22) by assuming that a perfective aspect appears in the structure beneath PERF (Iatridou et al., 2001; Pancheva, 2003). A lexical entry for PRFV can be found in (5a). (5a) expresses a relation where the RT contains the ET. While I return to the semantics of the perfective in greater depth in Section 3.2.2, for the purposes of the current example, what matters is that when PRFV is under the perfect, it expresses that the PTS contains the ET. This results in an experiential reading.

[5] While I follow a weak-XN semantics (Pancheva & von Stechow, 2004), the perfect has also been analyzed using a strong-XN (Dowty, 1979), anteriority (Reichenbach, 1947; Klein, 1994), and result state semantics (e.g., Kamp & Reyle, 1993).

[6] The temporal relation provided by $\leq$ is the reverse of $\geq$ in (20a). Since the NON-PAST semantics mirrors the NON-FUTURE, the temporal relation invoked by a NON-FUTURE tense is therefore equivalent to that of the perfect aspect, as Matthewson (2006) points out.

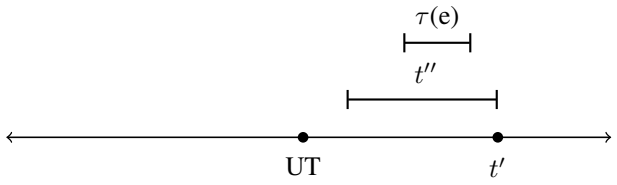
With this background on the perfect in mind, the hypothesized structure for a sentence like (13a) under a non-past tense analysis is shown in (23a), along with its predicted truth conditions in (23b). In (23a), NON-PAST appears in a position above PERF.

(23) a. [$_{TP}$ [$_{PerfP}$ [$_{AspP}$ [$_{vP}$ Tselmeg read poem] $_{Asp}$ PRFV] $_{Perf}$ PERF] $_{T}$ NON-PAST]

b. $[\![(23a)]\!]^{g,c}$ is true iff ∃t'[t' ≥ t_c & ∃t"[t" ≤ t' & ∃e[τ(e) ⊆ t" & read(e, Tselmeg, poem, w_c)]]]

According to the truth conditions in (23b), future-oriented readings of the perfect should be available if *-nA* contributes a non-past meaning. A future-oriented configuration that would satisfy the truth conditions in (23b) is visualized in (24). A future time *t'* serves as the RT and is the right boundary of *t"*, the PTS. *t"* extends from *t'* into the past and contains $\tau(e)$, the duration of a reading-a-poem event by Tselmeg. Each of *t'*, *t"*, and $\tau(e)$ is in the future, and none overlap with the UT. Yet, even though the non-past account predicts that a future perfect reading should be possible for the sentence (22), only a present perfect interpretation is available. Thus, the non-past tense semantics in (20) cannot successfully capture the aspectual data.

(24)



Further evidence against analyzing *-nA* as NON-PAST comes from sentences with conjoined subjects, based on the diagnostic from Matthewson (2006) to argue for an underspecified NON-FUT tense in St'át'imcets. The logic behind this diagnostic is as follows: for distributive readings of conjoined subjects, there is a sub-event that corresponds to each subject. For instance, in a sentence like "Mozart and Salieri slept," there is a sub-event of Mozart's sleeping and a sub-event of Salieri's sleeping. These sub-events need not overlap temporally, so long as they both occurred within the RT. In other words, one could utter "Mozart and Salieri slept" in a scenario where they both slept in the past, but at different times. If an underspecified tense like NON-FUT were to introduce an RT large enough to overlap with both the present and the past, it follows that NON-FUT should be acceptable in contexts where one sub-event is past-oriented and the other is present-oriented. In St'át'imcets, this is borne out (25).

(25) Context: Last year, John didn't go fishing, so he had no dried salmon last winter. Then summer came, and he went fishing. He got a lot of dried salmon. Fred didn't go fishing, so Fred has no dried salmon now.

(wa7) zúqw-cen s-John múta7 s-Fred
(IMPF) die-foot NOM-John and NOM-Fred
'John and Fred were/are starving.' (not at the same time) (from Matthewson, 2006)

Applying this diagnostic to Khalkha Mongolian, a NON-PAST semantics predicts that it should be possible to mark a verb with *-nA* to describe a scenario that has a non-overlapping (or,

only partially overlapping) present-oriented sub-event and future-oriented sub-event. However, as seen in (26), this is not the case. *-nA* is infelicitous when the temporal orientation of the two sub-events differs, shown in (26a), and is felicitous only when their temporal orientation is the same, like for the present context in (26b).[7]

(26) a. Context: Nyamdorj and Pürevsüren are interested in Icelandic. Nyamdorj already knows Icelandic fluently. Pürevsüren doesn't know any Icelandic but is planning to take an intensive class soon. He'll be fluent when the class ends.

#Nyamdorj, Pürevsüren khoyor Island khel med-**ne**.
Nyamdorj, Pürevsüren two Iceland language know-*nA*
Intended: 'Nyamdorj and Pürevsüren know/will know Icelandic.' (not at the same time)

b. Context: Nyamdorj and Pürevsüren are interested in Icelandic. Nyamdorj already knows Icelandic fluently, and so does Pürevsüren.

Nyamdorj, Pürevsüren khoyor Island khel med-**ne**.
Nyamdorj, Pürevsüren two Iceland language know-*nA*
'Nyamdorj and Pürevsüren know Icelandic.'

Taken together, the temporal reference of sentences with non-perfective aspects and conjoined subjects suggests that the behavior of *-nA* cannot be derived under a non-past tense analysis.

3.2.2 The present perfective paradox

To account for the observed distribution of present and future readings, I maintain the status of *-nA* as a tense by proposing that *-nA* should instead be treated as a present tense marker. My argument is as follows: if *-nA* is a present tense, then the lack of present-oriented readings for perfective sentences with eventive VPs can be attributed to the cross-linguistic incompatibility between the perfective aspect and the present tense (e.g., Bennett & Partee, 1972; de Wit, 2016). To illustrate the 'present perfective paradox' (Malchukov, 2009), semantic entries are given in (27) for PRFV, repeated from (5a), and PRS, repeated from (4).

(27) a. $[\![\text{PRFV}]\!]^{g,c} = \lambda P.\lambda t.\lambda w.\exists e[\ \tau(e) \subseteq t\ \&\ P(e)(w)\ ]$
b. $[\![\text{PRS}]\!]^{g,c} = \lambda p.\exists t'[\ t' = t_c\ \&\ p(t')(w_c)\ ]$

As previously discussed, the semantics of the perfective in (27a) states that the ET is located within a time *t*. In a present perfective sentence, *t* would be contributed by the present tense in (27b) and equivalent to the UT.

[7] For a future context, speakers preferred the sentence in (i), where *-nA* appears on the higher verb *bolokh* 'to become' and *med-* 'know' is inflected with the participial suffix *-kh* (see Section 5). It is unclear to me why the particular VP *Island khel med-* 'know Icelandic' exhibits this preference, differing from *minii nuutsiig med-* 'know my secret' in (1b).

(i) Context: A woman living in Iceland just gave birth to twins named Nyamdorj and Pürevsüren. She plans to raise them as Icelandic speakers.

Nyamdorj, Pürevsüren khoyor Island khel med-ekh bol-no.
Nyamdorj, Pürevsüren two Iceland language know-*kh* become-*nA*
'Nyamdorj and Pürevsüren will know Icelandic.'

What, then, makes the present tense incompatible with the perfective? The two accounts of 'simple present' sentences by Bennett & Partee (1972) and Ogihara (2007) provide different answers to this question. Under the explanation of Bennett & Partee (1972), the UT is an instantaneous, fleeting point in time.[8] Thus, for an interval that is equivalent to the UT to contain the duration of an eventuality, per the semantics of the perfective aspect in (27a), the eventuality would also need to be instantaneous.

In contrast, Ogihara (2007) defines the UT not as punctual, but as an interval. The duration of this interval is the time it takes the speaker to utter a given sentence. Under this non-instantaneous definition, however, the duration of the event can now be plausibly contained by the UT. Take, for example, an accomplishment like *read a postcard* or an activity like *dance*: both can be completed within the time it takes to say a sentence. As a result, having an RT that is equivalent to the UT is, itself, not inconsistent with the temporal relation encoded by the perfective in (27a). Instead, Ogihara (2007) traces the incompatibility of the present and the perfective to the grammatical rule in (28).

(28) For any simple present sentence ϕ, I is the entire interval needed to utter ϕ. ϕ uttered at I is true iff for all subintervals I' of I, $[\![\phi]\!]^{I'}$ is true, where $[\![\alpha]\!]^{I'}$ indicates for any expression α its extension at I'. (from Ogihara, 2007)

The rule in (28) is concerned with what it means for a sentence to be true of the UT. According to (28), the eventuality must hold throughout the UT interval in order for the sentence to be true, rather than holding *at* the UT. Only statives, which possess the subinterval property (Bennett & Partee, 1972), can satisfy this. A definition of the subinterval property is given in (29). Because the rule in (28) requires the sentence to hold at each subinterval of the UT, only a predicate with the subinterval property can guarantee the truth of the sentence at infinitesimally smaller subintervals.

(29) A predicate P possesses the subinterval property iff for every time interval I where P holds, P also holds at every subinterval I' of I. (from Bennett & Partee, 1972)

The interaction between the rule in (28) and the containment relation of the perfective is what rules out the unattested readings of eventive VPs. Further, even if the ET were equivalent in duration to the UT, as allowed by the non-proper subset relation in (27a), eventive VPs' lack of the subinterval property means that they are unable to hold throughout the UT (i.e., at each of its subintervals). With the rule in (28) not being satisfied, present perfective sentences are infelicitous with eventive VPs.

In sum, both Bennett & Partee (1972) and Ogihara (2007) predict the unacceptability of present perfective readings with eventive VPs. Across many languages, this prediction is substantiated: cross-linguistically, when a sentence possesses an eventive VP marked with present perfective morphology (i.e., 'simple present' morphology), other temporal readings arise in lieu of the infelicitous present perfective (de Wit, 2016). I propose that in Khalkha Mongolian, aspectually bare verbs marked with *-nA* constitute such a case of present perfective morphology. This approach makes explicit the descriptive comparisons drawn between sentences with *-nA* in Khalkha Mongolian and 'simple present' sentences in English (Binnick, 2016).

To understand how future readings obtain with *-nA*, I now discuss the attested temporal readings for sentences marked as present perfective across different languages. Languages

[8] Punctual characterizations of the UT are also given by Dowty (1979) and Giorgi & Pianesi (1997).

vary in what alternate interpretation arises in such sentences. For languages like English, sentences with present perfective morphology receive habitual readings, as shown in (30a) (e.g., Cable, 2020; Deo, 2015). It is infelicitous to use such sentences to describe actions that are currently ongoing, demonstrated by (30b), where the progressive is preferred over the unacceptable perfective.

(30) a. (Every day,) Maryam drinks matcha.
b. (Right now,) Maryam is drinking/#drinks matcha.

One analysis of how the habitual interpretation arises in (30a) is that such sentences are ambiguous between two structures: one structure contains a covert PRFV morpheme, whose incompatibility with the present renders it unacceptable, and the other structure contains a covert habitual operator (e.g., Cable, 2020). Examples of these structures are shown in (31) for the sentence in (30a). With the present perfective structure ruled out, only the present-oriented habitual reading arises.

(31) a. #[$_{TP}$ PRS [$_{AspP}$ PRFV [$_{vP}$ Maryam drink matcha]]]
b. [$_{TP}$ PRS [$_{AspP}$ HAB [$_{vP}$ Maryam drink matcha]]]

For other languages, such as Russian, Polish, and Slovenian, sentences with present perfective morphology instead result in future-oriented readings (e.g., Comrie, 1976; de Wit, 2016; Malchukov, 2009; Rivero, 2023). For instance, when a verb is marked with both present and perfective morphology in Russian (32), the eventuality it describes is interpreted as occurring in the future, rather than at the present. Given the future readings of sentences like (1), Khalkha Mongolian thus patterns like Russian concerning the present perfective paradox.

(32) On pri-det.
he at.PRFV-come.PRS.3SG
'He will come.' (from Malchukov, 2009)

If we assume that these future readings result from a structural ambiguity, analogous to (31) for the English simple present, the following is expected: (i) an infelicitous structure which contains both PRS and PRFV, and (ii) a felicitous structure which contains covert element(s) that introduce future-shifting. In the following section, I present an account of future shifting with *-nA* in Khalkha Mongolian which utilizes these structures.

4 A formal analysis of temporal reference with *-nA*

Based on how *-nA* interacts with the Aktionsarten and aspect, I have argued that it is a present tense marker. However, this raises a new question: if *-nA* is strictly present tense, then what contributes the future shifting in sentences like (1)? To answer this, I claim that futurity is introduced not by *-nA* itself, but by a covert modal and covert prospective aspect. I first summarize a decompositional approach to the semantics of the future (Section 4.1). Section 4.2 then outlines the analysis of future reference with *-nA*.

4.1 The semantic ingredients for future shifting

An emerging body of work on the semantics of futures has proposed that future reference can be decomposed into distinct aspectual and modal morphemes, where the former introduces temporal future shifting and the latter introduces quantification over possible worlds (see Bochnak, 2019). For example, consider English sentences that express future temporal reference with *will* (e.g., "Mozart will play the harpsichord"). In these proposals, *will* is treated as a tensed form of the modal morpheme WOLL (Abusch, 1997). A toy semantics for WOLL under a decompositional approach is given in (33), where WOLL is a modal that does not introduce any future RT itself (Kratzer, 2011; Mucha, 2016; Rivero, 2023).

(33) $[\![\text{WOLL}]\!]^{g,c} = \lambda p.\lambda t.\lambda w.\forall w' \in \text{Acc}(w,t)[\ p(t)(w')\ ]$ (adapted from Bochnak, 2019)

Because this modal is temporally unspecified, the forward temporal shifting associated with future expressions must have another source. This is accomplished via a prospective aspect, given in (34). PROSP quantifies over times, introducing a time *t'* in the future relative to the time *t* from tense. The modal in (33) takes scope over the prospective (Kratzer, 2011; Mucha, 2016).

(34) $[\![\text{PROSP}]\!]^{g,c} = \lambda p.\lambda t.\lambda w.\exists t'[\ t < t' \ \&\ p(t')(w)\ ]$

This semantics treats the prospective as a type of high aspect, rather than viewpoint aspect (Pancheva & Zubizarreta, 2023). As discussed before, high aspects like the prospective (34) and the perfect (21) establish a temporal relation between the RT provided by tense and an additional RT, unlike viewpoint aspects which relate an RT and ET.

A consequence of analyzing the prospective as a high aspect is that it cannot encode a temporal relation with the ET on its own. Thus, it is necessary for viewpoint aspect to appear in a lower position in order to relate the RT from high aspect to the ET. Support for such a role of viewpoint aspect can be found for both the perfect and prospective. Perfect participle morphology (e.g., the perfective *-ed*/the progressive *-ing* in English) has been analyzed as the realization of viewpoint aspect, resulting in different perfect readings (Iatridou et al., 2001; Pancheva, 2003). Likewise, viewpoint aspect can also derive different future readings (Copley, 2002; Mucha, 2016; Rivero, 2023). Verbs beneath English *will* can be marked with *-ing* for a progressive reading and unmarked for a perfective or habitual reading (Mucha, 2016).

To recapitulate, the decompositional approach outlined here necessitates two separate morphemes for future shifting. Given this, Mucha (2016) proposes that languages may be parameterized regarding which of these elements surfaces in future-marked sentences. That is, languages are predicted to vary based on whether the modal, the prospective aspect, or both surface overtly. For example, a language can achieve future reference via a covert modal and an overtly marked prospective aspect, as with *dim* in Gitksan (Matthewson et al., 2022) and *ta* in Paraguayan Guaraní (Tonhauser, 2011; Pancheva & Zubizarreta, 2023).[9] Other languages are analyzed as only realizing the modal morpheme overtly, such as with English WOLL (Mucha, 2016; Kratzer, 2011) and Medumba *á'* (Mucha, 2016), and the prospective appears covertly instead. Finally, both morphemes can surface: Mucha (2013) argues that future marking in

[9] As predicted by the decompositional view, a modal and prospective can simultaneously be overt in Gitksan, realized as two separate morphemes (Matthewson et al., 2022). Future reference is possible when the prospective *dim* appears without an overt modal, but *dim* can also co-occur with the circumstantial possibility modal *da'akhlxw*.

Hausa overtly realizes both the modal and prospective aspect. This is evidenced by the future-shifting requirements of the modal *zā* and its obligatory co-occurence with a low tone on weak subject pronouns, which are marked with TAM morphology in Hausa. Mucha (2013) analyzes this low tone as a prospective aspect.

Altogether, the future-marking typology described above includes languages of three types: (i) those with a covert MOD and overt PROSP, (ii) those with an overt MOD and covert PROSP, and (iii) those with an overt MOD and overt PROSP. However, another logically possible type of language is predicted by this typology, though previously unattested: a language where future shifting is achieved when MOD and PROSP are *both* unpronounced.

I argue that Khalkha Mongolian provides an example of such a language. Thus, for future-oriented sentences with *-nA*, Khalkha Mongolian expresses futurity with two morphemes: a prospective aspect, which is covertly present in a high aspectual position, and a modal, which appears above the prospective. While the presence of a covert modal is supported by the licensing of modal subordination with *-nA* (Section 3.1.4), I provide explicit argumentation for the presence of these covert operators in Section 5.

4.2 Deriving the temporal reference of *-nA*

To account for the distribution of present and future reference for verbs marked with *-nA*, I analyze these different readings as arising from an ambiguity between two structures. These structures are shown in (35), repeated from (2).

(35) a. Future-oriented structure:
$[_{TP}\ [_{ModP}\ [_{AspP}\ [_{AspP}\ [_{vP}\ \dots]\ _{Asp}\ \textsc{prfv}]\ _{Asp}\ \textsc{prosp}]\ _{Mod}\ \textsc{mod}]\ _{T}\ \textsc{prs}]$

b. Present-oriented structure:
$[_{TP}\ [_{AspP}\ [_{vP}\ \dots]\ _{Asp}\ \textsc{prfv}]\ _{T}\ \textsc{prs}]$

In Section 3.2, I argued for the status of *-nA* as a present tense, rather than non-past. The semantics for PRS are repeated in (36). As discussed before, the present tense provides an RT that is equivalent to the UT.

(36) $[\![\textit{-nA}]\!]^{g,c} = [\![\textsc{prs}]\!]^{g,c} = \lambda p.\exists t'[\ t' = t_c\ \&\ p(t')(w_c)\ ]$

In the absence of a non-past tense, the semantic contribution of the present tense *-nA* on its own is not sufficient for future reference. The source of futurity is the combination of two operators: a modal and a prospective aspect, both of which appear covertly alongside *-nA*.

A preliminary semantics for the modal operator is given in (37). The appropriate modal base for future expressions remains an unsettled topic, with proposed modal bases for futures including metaphysical (Condoravdi, 2002), circumstantial (Abusch, 2012), and epistemic (Giannakidou & Mari, 2018). Disambiguating between these proposals is outside the scope of the current paper. I assume a circumstantial modal base for the relevant cases, following Abusch (2012), and a stereotypical ordering source.

(37) **Preliminary version:**
$[\![\textsc{mod}_{\textsc{circ}}]\!]^{g,c} = \lambda p.\lambda t.\lambda w.\forall w' \in \textsc{best}_{h(w,t)}(\bigcap \textsc{mb}_{\textsc{circ}}(w,t))[\ p(t)(w')\ ]$

Because (37) lacks temporal specification, a temporal component, specifically a null prospective aspect, is needed to introduce a future RT. A semantics for PROSP is given in (38), where PROSP introduces an RT which temporally succeeds the RT from tense.

(38) $[\![\text{PROSP}_i]\!]^{g,c} = \lambda p.\lambda t.\lambda w.\exists t'\ [\ t < t'\ \&\ t' \subseteq g(i)\ \&\ p(t')(w)\ ]$

In (38), quantification over future times is restricted by *g(i)*, which picks out a time interval that is salient in the context of the utterance. An empirical reason to restrict quantification comes from examples like (39): the truth conditions of these sentences would not be met with *any* future time at which Uyanga drank water. With unrestricted quantification, such sentences would only be false in a limited number of scenarios, as drinking water is necessary for Uyanga to live. Because the prediction in (39) only concerns whether Uyanga will drink water during the dinner party, the future time introduced by PROSP needs to be restricted accordingly.

(39) Context: You've invited your friends to a dinner party. Uyanga usually prefers to drink wine, but you recently learned that she's pregnant. You're talking about what she'll drink at the party instead.
   a. Uyanga will drink water.
   b. Uyanga us uu-**na**.
      Uyanga water drink-*nA*
      'Uyanga will drink water.'

Following cross-linguistic work on the future (see Section 4.1), the prospective aspect is treated as a high aspect in (38). I analyze this morpheme as occurring above a covert perfective aspect. However, while the imperfective should also be able to appear under PROSP, Section 3.1.3 illustrated that this is not possible in Khalkha Mongolian: the relevant future-oriented structure is only available with the perfective. I propose that this is due to a selectional restriction imposed by PROSP, where it selects for PRFV. This claim is supported by how future-shifting is accomplished for the imperfective (see 9b, 12a): a higher verb *bolokh* 'to become' is required, to which PRFV can attach. This claim is also supported by language-specific selectional restrictions attested for high aspects elsewhere: the Greek perfect, for example, requires a perfective viewpoint aspect (Iatridou et al., 2001).

I will now show how these semantic components come together to derive the meaning of eventive VPs (Section 4.2.1) and stative VPs (Section 4.2.2) marked with *-nA*.

4.2.1 Eventives

Consider the sentence in (40), where the eventive VP *bich-* 'write' is inflected with *-nA*. This sentence can only receive a reading where Margad writes a letter at a future time, rather than writing the letter at the UT.

(40) Margad (margaash) zakhia bich-**ne**.
   Margad (tomorrow) letter write-*nA*
   'Margad *writes/will write a letter (tomorrow).'

Under this proposal, there are two possible structures for the sentence in (40), both of which are given in (41). Due to the incompatibility between the present tense and perfective aspect (Section 3.2.2), the structure in (41b) is infelicitous.

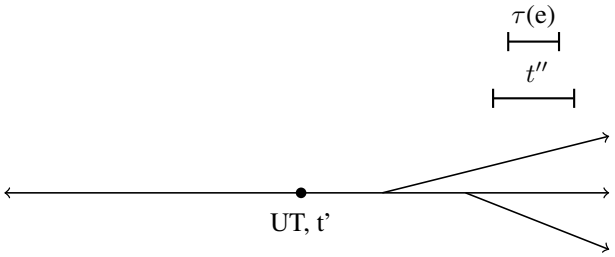
(41) a. $[_{\text{TP}}\ [_{\text{ModP}}\ [_{\text{AspP}}\ [_{\text{AspP}}\ [_{\text{vP}}$ Margad write letter$]_{\text{Asp}}$ PRFV$]_{\text{Asp}}$ PROSP$]_{\text{Mod}}$ MOD$]_{\text{T}}$ PRS$]$
   b. #$[_{\text{TP}}\ [_{\text{AspP}}\ [_{\text{vP}}$ Margad write letter$]_{\text{Asp}}$ PRFV$]_{\text{T}}$ PRS$]$

The future-oriented structure in (41a), on the other hand, does not result in infelicity due to the semantic contribution of PROSP. Consider the truth conditions of (41a), shown in (42): the future time t'' introduced by PROSP is what contains the duration of the eventuality $\tau(e)$, rather than a time that is equivalent to the UT. Because of this, no account of the present perfective paradox rules out this LF. Under a Bennett & Partee (1972) view, the truth conditions are satisfiable since t" is an interval and not a point, allowing it to contain the ET. For Ogihara (2007), the new RT t" doesn't conflict with the rule in (28).

(42) $[\![(41a)_i]\!]^{g,c,h}$ is true iff ∃t'[t' = t_c & ∀w' ∈ $\text{BEST}_{h(w_c,t')}(\bigcap \text{MB}_{\text{CIRC}}(w_c,t'))$[∃t"[t' < t" & t" ⊆ g(i) & ∃e[τ(e) ⊆ t" & write(e, Margad, letter, t", w')]]]]

A configuration that verifies the truth conditions in (42) is displayed in (43). Here, the branches extending from the timeline are a visual representation of different possible worlds: the actual world, and those worlds which diverge from the actual world.

(43)



4.2.2 Statives

I argued that with eventive VPs, there is an ambiguity between two underlying structures. Both of these structures contain the perfective at a low AspP, but one is rendered infelicitous due to the present perfective paradox. Turning to statives, a structural ambiguity with *-nA* is also present, resulting in the availability of both present- and future-oriented readings like in (1b). However, because statives are traditionally thought to be incompatible with the perfective (Bary, 2009; de Swart, 1998; Dieuleveut, 2023; Mari & Martin, 2007), it is unclear whether the same pair of structures underlies the ambiguity for both eventives and statives. In order to derive the temporal ambiguity with stative VPs, then, a question first needs to be answered: what aspectual elements are present when stative VPs are marked with *-nA*? I give background on the interaction between perfectivity and statives before presenting the analysis, where I treat statives inflected with *-nA* as perfective.

In languages like French (Dieuleveut, 2023; Mari & Martin, 2007) where perfectivity is overtly marked, statives are unacceptable with the perfective. This incompatibility is thought to arise from a boundedness requirement of the perfective, which necessitates that the RT contains both the start and end of the eventuality (de Swart, 1998; Dieuleveut, 2023; Mari & Martin, 2007). Stative VPs, unlike eventive VPs, lack a definitive start or end point and are thus unbounded (e.g., Altshuler & Schwarzschild, 2013). Because the unboundedness of statives is at odds with the perfective's boundedness requirement, the two categories are predicted to be incompatible.

To adhere to the boundedness requirement in languages like English, where an absence of overt aspectual morphology results in perfective readings, statives are typically treated as appearing with a covert imperfective aspect (Deo, 2009; Dowty, 1979).[10] Since English stative VPs like in (44a) cannot be marked with overt progressive morphology (Dowty, 1975; Lakoff, 1966), their continuous meaning is instead traced to the unpronounced imperfective. However, unlike English, stative VPs in Khalkha Mongolian can be overtly marked as imperfective (44b), repeated from (14b), or as habitual (44c). If aspectually bare statives in Khalkha Mongolian involved covert imperfective morphology, as has been proposed for English, it would be necessary to explain why the imperfective can be optionally realized as either overt or covert. It would also be necessary to account for why the same optionality does not hold for eventive VPs, which can only be interpreted as imperfective with imperfective morphology on the surface.

(44) a. #The professor is knowing my name.

b. Bat minii nuuts-iig **med-ej bai-na**.
Bat 1SG.POSS secret-ACC know-CVB AUX-*nA*
Literal: 'Bat is knowing my secret.'

c. Suvdmaa Frants khel **med-deg**.
Suvdmaa France language know-HAB
Literal: 'Suvdmaa (habitually) knows French.'

It has also been debated whether statives like *know my secret* in (44b) lack an eventuality argument entirely, either because of their status as Kimian states (Maienborn, 2007), which by definition cannot be located spatiotemporally, or simply by virtue of being stative in the first place (Katz, 2003). Yet, in Khalkha Mongolian, such statives can still be contrastively marked for viewpoint aspect (44). Given the Neo-Reichenbachian approach to viewpoint aspect assumed here, such marking requires an eventuality argument whose time holds a relation with the RT. These contrasts thus cannot be analyzed without the presence of an eventuality argument.

Where does this leave us regarding the perfective aspect and statives? Recent work on the semantics of the perfect has independently proposed the existence of perfective statives to derive the availability of experiential readings of the perfect with statives (e.g., von Fintel & Iatridou, 2019; Rouillard, 2026). When statives appear with the perfect, experiential and universal readings are both available. With an experiential reading, like (45), it is inferred that Colleen's tiredness ended before the UT.

(45) Colleen has been tired before, (but she isn't now). (Experiential)

If the stative in (45) were imperfective, a reading where the eventuality holds only at a subset of the PTS should not be available.[11] This is because the temporal relation encoded by the imperfective (see 5b) would lead (45) to entail that the ET contains the PTS. Since the

[10] In Kamp & Reyle (1993), the relationship between the (im)perfective and stativity is captured under a DRT approach. Rather than covert morphemes, temporal location conditions distinguish between eventive/stative VPs: the condition for eventives states that for an event e and location time t, $\tau(e) \subseteq t$, while statives require that for a state s and location time t, $t \subseteq \tau(s)$. Under this approach, the Aktionsart of the VP, rather than viewpoint aspect, drives the distinction.

[11] See Pancheva (2003) for an analysis of the experiential readings of statives which instead uses a 'neutral imperfective' (Smith, 1991).

PTS for the present perfect overlaps with the UT (see Section 3.2.1), the ET would therefore contain the UT. However, this prediction is at odds with the actual meaning of the experiential in (45), where the being-tired eventuality need not hold anymore.

In light of these observations, there is not sufficient evidence to rule out these stative VPs being aspectually perfective in Khalkha Mongolian. I therefore give an alternative analysis to account for the viewpoint aspect of these statives: like the eventive VPs, aspectually bare stative VPs appear alongside a covert perfective. I argue that the temporal meaning of statives marked with *-nA* can, too, be derived via the pair of structures in (35), just like the eventives. The implementation of this account centers around two claims: (i) because statives have the subinterval property (Bennett & Partee, 1972), a subinterval of the state itself may satisfy the perfective's semantics, and (ii) per Ogihara (2007), a grammatical rule is active that requires eventualities overlapping with the UT interval to hold throughout the UT, which only statives can satisfy.

The subinterval property states that if the eventuality is true at some interval *I*, then it is also true at every subinterval of *I* (Bennett & Partee, 1972). Thus, because stative VPs possess the subinterval property, they are true not only for the maximal interval they hold over, but also for each of its subintervals. This property of stative VPs allows them to satisfy the semantics of the perfective in (27a). The existential quantification over eventualities in (27a) requires only that there is at least one eventuality whose duration is either a subset of or equal to the RT. For a state *s*, one of its substates s_{sub} would be able to verify this temporal relation imposed by the perfective: even if the maximal interval that *s* holds over contains the RT, a sufficiently small subinterval (i.e., the duration of s_{sub}) that is equivalent in duration to the RT would satisfy (27a). Therefore, regardless of whether the UT is treated as a point (e.g., Bennett & Partee, 1972) or as an interval (e.g., Ogihara, 2007), the subinterval property results in the felicity of perfective statives in the present tense.

That being said, I opt to follow the account of the present perfective paradox in Ogihara (2007) rather than the original proposal of Bennett & Partee (1972) in order to capture the behavior of achievement VPs with the present tense. Despite being punctual, achievements like *become* in (46) are unable to receive a present perfective reading (Dieuleveut, 2023). This is puzzling if the punctuality of the UT is the source of the present perfective paradox, as the duration of the achievements in (46) should be equivalent to the UT and thus satisfy the semantics of the perfective in (27a).

(46) a. Nomin bagsh bol-**no**.
Nomin teacher become-*nA*
'Nomin *becomes/will become a teacher.'

b. Eli becomes happy (whenever I see him/#now).

However, assuming a grammatical rule like (28), the lack of present perfective readings for achievements can be accounted for. Achievements, being punctual, are unable to hold throughout the interval of the UT, resulting in their infelicity with present perfective sentences. Statives, on the other hand, are able to hold throughout the UT interval, as discussed in Section 3.2.2. For this reason, I adopt the rule in (28). I'll now lay out the details of this account for the sentence in (47).

(47) Erdenechimeg shaltgaan-iig med-**ne**.
Erdenechimeg reason-ACC know-*nA*
'Erdenechimeg knows/will know the reason.'

Treating the stative examples as perfective, the relevant structures are displayed in (48). Like the eventive VPs, there is an ambiguity between a present-oriented and a future-oriented structure. However, both structures, including the present perfective in (48b), are felicitous for stative VPs. This leads to the availability of the present-oriented reading.

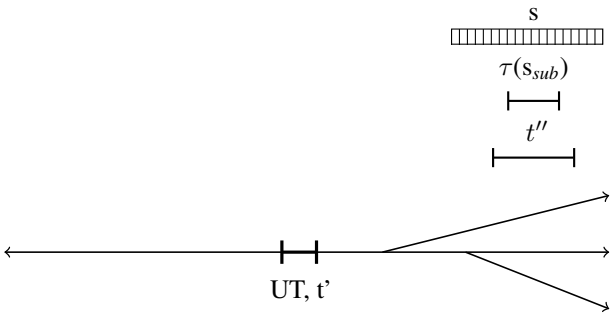
(48) a. [$_{TP}$ [$_{ModP}$ [$_{AspP}$ [$_{AspP}$ [$_{vP}$ Erdenechimeg know reason] $_{Asp}$ PRFV] $_{Asp}$ PROSP] $_{Mod}$ MOD] $_{T}$ PRS]
b. [$_{TP}$ [$_{AspP}$ [$_{vP}$ Erdenechimeg know reason] $_{Asp}$ PRFV] $_{T}$ PRS]

The truth conditions for the stative sentences are shown in (49), with the future-oriented reading in (49a) and the present-oriented reading in (49b). In (49a), the future time *t"* from PROSP contains the duration of a state *s*, but in (49b), where both MOD and PROSP are absent, a time that is equivalent to the UT contains the duration of *s* instead.

(49) a. $[\![(48a)_i]\!]^{g,c,h}$ is true iff $\exists$t'[t' = t_c & $\forall$w' $\in$ $\text{BEST}_{h(w_c,t')}(\bigcap \text{MB}_{\text{CIRC}}(w_c,t'))$[$\exists$t"[$\exists$s[τ(s) $\subseteq$ t" & t' $<$ t" & t" $\subseteq$ g(i) & know(s, Erdenechimeg, reason, t", w')]]]]
b. $[\![(48b)]\!]^{g,c}$ is true iff $\exists$t'[t' = t_c & $\exists$s[τ(s) $\subseteq$ t' & know(s, Erdenechimeg, reason, t', w_c)]]

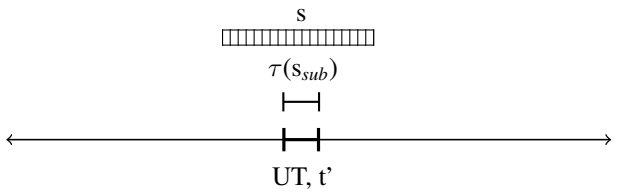
A configuration is given in (50) that verifies the future-oriented truth conditions in (49a). Though *s* itself extends past the bounds of *t"*, there still exists a substate of *s*, s_{sub}, whose duration is contained within *t"*. Therefore, the configuration in (50) renders (49a) true.

(50)



A configuration that verifies the present-oriented truth conditions in (49b) is displayed in (51). Once again, the duration of *s* contains *t'* (i.e., the UT). Regardless, however, there still exists an s_{sub} that is equivalent in duration to *t'*. Because PRFV in (27a) simply requires the existence of an eventuality whose duration is contained by or equivalent to *t'*, this is satisfied via s_{sub}.

(51)



To summarize, this analysis treats the morpheme *-nA* as a present tense, with future readings arising as a result of a null modal and a null prospective. Present-oriented readings obtain for statives, but not eventives, due to the present perfective paradox: statives possess the subinterval property, allowing them to satisfy the grammatical rule in (28), whereas eventives do not. A theoretical consequence of this analysis thus bears on the relationship between stativity and perfectivity. To what degree does the range of aspectual categories that can appear with stative VPs vary cross-linguistically? In Section 4.2.2, I modeled the relevant stative VPs as co-occurring with the perfective aspect in Khalkha Mongolian. Taken together with their ability to appear with imperfective and habitual morphology in (44b, 44c), this analysis treats Khalkha Mongolian as a language where stative VPs can combine more freely with different viewpoint aspects.

5 Covert modality and *-kh*

A decompositional account of future reference in Khalkha Mongolian relies on two null operators: a modal, and a prospective. As evidence for these covert operators, I argue that despite being covert in affirmative sentences with *-nA*, a modal suffix *-kh* surfaces under sentential negation (Section 5.1). This pattern is traced to morphosyntactic restrictions on TAM marking and negation. I then probe the status of *-kh* as a modal in Section 5.2, showing that it patterns like other modal elements cross-linguistically. A separate covert prospective is supported by present readings of the modal *-kh* (Section 5.2.1).

5.1 Interactions between negation and *-nA*

Sentential negation in Khalkha Mongolian involves morphological alternations that indicate the presence of a modal under *-nA* in cases of future reference. I illustrate these points by overviewing how the morphosyntax of negation interacts with tense and aspect. Namely, verbs often bear different inflectional morphology in the presence of sentential negation than in their affirmative counterparts (Brosig, 2015; Svantesson, 1991).

As an initial observation, sentential negation is achieved through the negator *-güi*, which is involved in verbal inflection alongside tense and aspect. However, it is ungrammatical for *-güi* to co-occur with *-nA*. While the affirmative sentence with *-nA* in (52a), repeated from (40), is grammatical, (52b) is ungrammatical, regardless of whether tense appears in a higher or lower position than negation.[12]

[12] While Lim (2025) reports that examples like (52b) with tense above *-güi* are possible in colloquial speech, none of the consulted Khalkha Mongolian speakers accepted such sentences.

(52) a. Margad (margaash) zakhia bich-**ne**.
Margad (tomorrow) letter write-*nA*
'Margad will write a letter (tomorrow).'

b. *Margad (margaash) zakhia {bich-**ne-güi** / bich-**güi-ne**}.
Margad (tomorrow) letter {write-*nA*-NEG / write-NEG-*nA*}
Intended: 'Margad won't write a letter (tomorrow).'

Instead, negation must follow the suffix *-kh*. (53) exemplifies this with an eventive VP. Despite their morphological asymmetries on the surface, (53) maintains an interpretive relationship with (52a) (i.e., p vs. $\neg p$).

(53) Margad (margaash) zakhia bich-**ikh-güi**.
Margad (tomorrow) letter write-*kh*-NEG
'Margad won't write a letter (tomorrow).'

Parallel examples with a stative VP are given in (54). Like the affirmative example in (54a), repeated from (47), the negated stative in (54b) can be interpreted either as present- or future-oriented.[13]

(54) a. Erdenechimeg shaltgaan-iig med-**ne**.
Erdenechimeg reason-ACC know-*nA*
'Erdenechimeg knows/will know the reason.'

b. Erdenechimeg shaltgaan-iig med-**ekh-güi**.
Erdenechimeg reason-ACC know-*kh*-NEG
'Erdenechimeg doesn't/won't know the reason.'

To derive why *-nA* cannot appear with negation, I establish the following generalizations in this section: in Khalkha Mongolian, only a single TAM morpheme may surface directly on a verbal root alongside the negator. This restriction stems from two factors: verbal roots are rendered ungrammatical without TAM morphology, and, at the same time, a morphosyntactic restriction prohibits multiple overt TAM morphemes from inflecting. Further, the particular TAM morpheme which can surface is restricted to suffixes which form verbal participles. Because *-nA* is a tense morpheme, I conclude that it is too high syntactically to co-occur with the negator.

To walk through these generalizations, I start via the examples in (55) with the habitual suffix *-dAg*. In the affirmative sentence in (55a), the habitual marker appears directly above the verbal root *tsewerle-* 'clean'. When this sentence is negated in (55b), *-güi* attaches above the habitual marker. It is ungrammatical for *-güi* to appear directly on a verbal root, whether

[13]For some speakers, the future-oriented reading of (54b) is only possible when a converbial form of *olokh* "to find" appears alongside *med-* "know" (i), instead resulting in an eventive interpretation. Why is a future reading of (54b) not uniformly available? One alternative analysis is that the future reading of *med-* in (54b) was made available (for those speakers who accept it) via aspectual coercion, rather than involving a genuine stative. However, even if this were the case, the key contrast between statives and eventives would still hold: the present-oriented structure is only available with statives, not eventives.

(i) Erdenechimeg shaltgaan-iig ol-j med-ekh-güi.
Erdenechimeg reason-ACC find-CVB know-*kh*-NEG
'Erdenechimeg won't find out the reason.'

or not it bears *-dAg*. Thus, the overt presence of a verbal suffix like *-dAg* is a requirement for sentential negation to proceed, and the negator must attach higher than the suffix. Indeed, outside of negation, it is ungrammatical for a verbal root to appear without any sort of inflectional marking.

(55) a. Shinee ödör bür tsonkh-oo tsewerle-**deg**.
Shinee day every window-REFL.POSS clean-HAB
'Shinee cleans his windows every day.'

b. Shinee ödör bür tsonkh-oo {tsewerle-**deg-güi** / *tsewerle-**güi(-deg)**}.
Shinee day every window-REFL.POSS {clean-HAB-NEG / tsewerle-**güi(-deg)**}
'Shinee doesn't clean his windows every day.'

As shown by (55a–55b), tense is not phonologically realized on the surface when the verb is inflected with an overt aspectual suffix, regardless of the sentence's polarity. Yet, the only possible temporal interpretation of (55a–55b) is present-oriented. I suggest that the present tense possesses two allomorphs: *-nA*, and a covert form. In habitual examples like (55a–55b), the covert form appears. This stems from a morphosyntactic restriction against multiple overt TAM morphemes attaching to a single verbal root in Khalkha Mongolian.

Indeed, more complex TAM inflection (i.e., overtly realizing more than one TAM category) requires an additional auxiliary verb in the structure. This is shown in (56a) with the habitual suffix and the past tense, along with a negated form in (56b). The auxiliary is only necessary when there is an additional overt suffix to host. Khalkha Mongolian is not unique in this respect: similar behavior of verb periphrasis is attested in Kinande (Bjorkman, 2011) and Turkish (e.g., Kim, 2024).

(56) a. Shinee ödör bür tsonkh-oo tsewerle-**deg bai-saŋ**.
Shinee day every window-REFL.POSS clean-HAB AUX-PST
'Shinee cleaned his windows every day.'

b. Shinee ödör bür tsonkh-oo tsewerle-**deg-güi bai-saŋ**.
Shinee day every window-REFL.POSS clean-HAB-NEG AUX-PST
'Shinee didn't clean his windows every day.'

What happens when the verbal root is aspectually bare, like in future-shifted sentences with *-nA*? As seen in (52), *-nA* itself cannot co-occur with *-güi*. This asymmetry under negation arises because *-güi* is morphologically licensed only by aspectual suffixes that form verbal participles (Brosig, 2015). More specifically, this set of suffixes consists of four morphemes (alongside a descriptive label of the participle they appear in): *-dAg* (habitual participle), *-sAŋ* (past participle), *-(ɢ)AA* (imperfective participle), and *-kh* ("future" participle).

I propose that the incompatibility between *-nA* and sentential negation stems from the position that the negator *-güi* can occupy: as *-güi* can only attach to a verbal participle, it cannot attach above TP. I characterize the NegP projection that *-güi* appears at as a position above AspP, but below TP, depicted in (57) (see Lim, 2025). Therefore, in the relevant sentences, *-güi* surfaces with *-kh* and not *-nA*.[14]

[14]The structure in (57) places *-güi* above vP, the locus of existential closure over events (Diesing, 1992). Because sentential negation is typically thought to negate quantification over events, it's unsurprising that *-güi* would occupy such a position (Acquaviva, 1997; Zeijlstra, 2022).

(57) $[_{TP}\ [_{NegP}\ [_{AspP}\ [_{vP}\ \ldots]\ \text{Asp}\ ]\ \text{Neg}\ ]\ \text{T}\ ]$

To maintain an interpretive symmetry between affirmative sentences with *-nA* and their negative counterparts, it is necessary that the semantic contribution of the verbal participle inflected with *-kh* parallels that of verbs inflected with *-nA*. To unpack the semantics of the *-kh* morpheme, I next explore its distribution and argue for its status as modal.

5.2 A family resemblance with modals

The previous section demonstrated that when affirmative sentences with *-nA* are negated, the morpheme *-kh* surfaces. In the Mongolic literature, *-kh* has been descriptively labeled as an irrealis mood (Song, 1997; Svantesson, 1991) or an infinitival/future (Binnick, 2012). Drawing from these characterizations, I propose that *-kh* is a modal element, not simply an allomorph of *-nA* under negation. I present six data points showing that *-kh* possesses a family resemblance with other modals and appears in contexts which implicate modality.

First, *-kh* is required in counterfactual conditionals. Specifically, the *-kh* morpheme must appear in the consequent, inflected on a verb beneath a perfect auxiliary. (58a) is interpreted as a counterfactual, as opposed to an indicative conditional like (58b).

(58) a. Uyanga shükher-ee mart-saŋ bol ter boroo-nd nor-**okh** bai-saŋ.
Uyanga umbrella-REFL.POSS forget-PST COND 3.SG rain-DAT get.wet-*kh* AUX-PST
'If Uyanga had forgotten her umbrella, she would have gotten wet in the rain.'

b. Uyanga shükher-ee mart-saŋ bol ter boroo-nd {nor-**no** / nor-**soŋ**}.
Uyanga umbrella-REFL.POSS forget-PST COND 3.SG rain-DAT {get.wet-*nA* / get.wet-PST}
'If Uyanga forgot her umbrella, she {will get / got} wet in the rain.'

The finding in (58) is expected if *-kh* has a similar semantics to English WOLL. For conditionals in English, as well as Modern Greek, a counterfactual interpretation is only possible with a WOLL-like morpheme in its consequent (Iatridou, 2000). English examples are illustrated in (59), where a present counterfactual in (59a) is contrasted with an indicative conditional in (59b).

(59) a. If Mozart had composed an opera in Russian, Salieri would have attended one of its performances.

b. If Mozart composed an opera in Russian, Salieri attended one of its performances.

The presence of WOLL in the consequent of counterfactuals is commonly thought to introduce quantification over historically possible worlds, which stem from the past RT introduced in the antecedent (Arregui, 2007; Ippolito, 2003; Khoo, 2017). In (59), the past tense morphology on the verb *compose* sets the RT as a time *t* that precedes the time of speech. Under a Thomason-style framework, utilizing $T \times W$-frames, a pair of worlds *w* and *w'* are historical alternatives at the past RT *t* if they are equivalent at *t* and at all times preceding *t* (Thomason, 1970; Kaufmann, 2005). Put differently, *w* and *w'* are historical alternatives if they share a history up until *t*. However, at any *t'* such that $t < t'$, it is possible for *w* and *w'* to diverge from each other. As a consequence, alternative futures are thus accessible from *t* which differ from the state of affairs in the actual world at the UT. This quantification over historically possible worlds is, under this view, made possible by future-oriented, modal morphology like WOLL.

Second, the height of *-kh* determines its interpretation. When *-kh* appears in a high syntactic position on an auxiliary, it gives rise to an epistemic reading, as in (60).

(60) Context: Every day at noon, Enkhjin takes a break from work to go on a walk. It's noon now.

Enkhjin alkha-j bai-GAA bai-**kh**.
Enkhjin walk-CVB AUX-IPF AUX-*kh*
'Enkhjin must be walking.' (Epistemic)

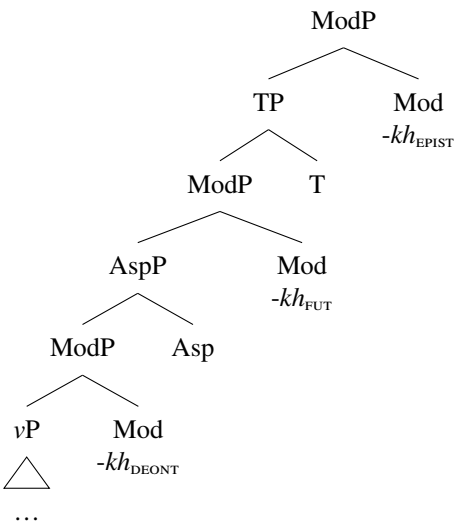
When *-kh* appears on a participle (i.e., in a low position), it behaves like a root modal. In sentences with deontic necessity interpretations like (61), *-kh* attaches to the verb and appears underneath the phrase *yos-toi* (lit. 'with the rules'). Considering (61) alongside the interpretation of *-kh* under negation in (53) and in counterfactuals in (59–66), it is clear that the modal flavor of *-kh* in low positions is not epistemic, but rather, it must be underspecified.

(61) Context: You and your friends, Sarangerel and Khulan, are going to an art museum. The museum requires tickets to enter the new exhibit. Sarangerel hasn't bought her ticket yet, and you're reminding Khulan what Sarangerel needs to do to see the exhibit.

Sarangerel bilet av-**akh yos-toi**.
Sarangerel ticket buy-*kh* rule-with
'Sarangerel must purchase a ticket.' (Deontic)

The effect of syntactic height is captured if *-kh* is a modal. Cross-linguistically, epistemic modals have been argued to scope above tense, while root modals like the deontic *-kh* scope underneath tense (e.g., Hacquard, 2009). The specific position occupied by root modals is analyzed variably in the literature, treated either as beneath aspect (Hacquard, 2009, 2010), or, for modals involved in future expressions, as above aspect (Kush, 2011; Shah, 2026). The latter is overtly exemplified in Khalkha Mongolian by *-kh* in sentences like (53). These differences in the relative height of *-kh* are displayed (62). The way positional differences between modals with the same form result in epistemic, future, and root interpretations has been variably derived (see, e.g., Hacquard, 2009, 2010; Kush, 2011; Shah, 2026). I set aside precisely how the position of *-kh* alters its modal flavor, as well as the restriction of *-kh* to verbal participles and the aforementioned auxiliary.

(62)



Third, *-kh* can trigger modal subordination (Roberts, 1989). Modal subordination is only licensed if a modal operator is present in each of the sentences in the relevant discourse. In the earlier modal subordination example (16b), the presence of *-kh* on a higher auxiliary makes the sentence acceptable. Thus, this suggests that *-kh* has a modalized semantics.

Fourth, negation is unable to scope above *-kh* when it is in a high position, shown in (63).[15] This is consistent with epistemic modals in other languages which cannot be outscoped by quantifiers, including negation (von Fintel & Iatridou, 2003).[16] Thus, examples like (63) provide additional evidence that when an auxiliary is inflected with *-kh* (rather than with a tense like *-nA*), it acts as an epistemic modal.

(63) Context: Enkhjin is your lazy coworker. You know that every day at noon, Enkhjin takes a break from work to take a nap. It's noon now.

a. *Enkhjin alkha-j bai-ɢaa bai-**kh**-güi.
Enkhjin walk-CVB AUX-IPF AUX-*kh*-NEG
Intended: 'Enkhjin must not be walking.' (Epistemic)

b. Enkhjin alkha-j bai-**kh**-güi bai-**kh**.
Enkhjin walk-CVB AUX-*kh*-NEG AUX-*kh*
'Enkhjin must not be walking.' (Epistemic)

Fifth, the 'acquaintance inference' (AI) of predicates of personal taste (PPTs) is obviated under *-kh*, like with other modals (e.g., Pearson, 2013; Klecha, 2014). When PPTs appear in the simple present, they give rise to an AI: the speaker is inferred to have first-hand experience that the PPT holds of the object. If a speaker were to utter a sentence like "Listening to Mozart is fun," one would assume that they had listened to Mozart before. However, when PPTs appear

[15]The consulted Khalkha Mongolian speakers varied in whether they accepted sentences like (63b), with negation underneath the epistemic modal. Crucially, however, all speakers judged sentences with negation above the epistemic modal as ungrammatical.

[16]Note that in (63b), *-kh* appears twice: the higher *-kh* is the epistemic modal, whereas the lower *-kh* forms a 'future' participle and can scope beneath negation.

under modals, this AI is obviated. If a speaker instead stated that "Listening to Salieri must be fun," no inference would arise that they had listened to Salieri's music beforehand.

AIs are indeed obviated under *-kh* in (64a), but they remain under present-oriented interpretations of *-nA* like (64b), where the *-nA*-marked auxiliary is optionally pronounced.[17] The obviation in (64a) favors a modal treatment of *-kh*.

(64) Context: Your friend just made *süütei tsai*, a salted tea drink. You've never tried their *süütei tsai* before. However, they've made tasty drinks for you before, and they always use high-quality ingredients.

a. Ene süütei tsai amttai bai-**kh** bai-**kh**.
this milky tea delicious AUX-*kh* AUX-*kh*
'This *süütei tsai* must be delicious.' (Epistemic)

b. #Ene süütei tsai amttai (bai-**na**).
this milky tea delicious AUX-*nA*
Intended: 'This *süütei tsai* is delicious.'

Sixth, *-kh* can appear on a high auxiliary in cases of abductive reasoning. Abductive reasoning involves drawing inferences about the cause of an eventuality from the eventuality itself (Peirce, 1955). For instance, say that your friend, who has always been blonde, suddenly has red hair. You would conclude that she dyed it. This is a case of abductive reasoning: it is inferred that the having-red-hair eventuality was caused by a hair-dyeing event.

In contrast, with deductive reasoning, inferences are made about the eventuality from its cause. Imagine that your friend calls you on the phone, telling you that she used red hair dye. Because you know that using hair dye leads to a change in hair color, you conclude that her hair is now red. This reasoning is non-abductive: it is inferred that the hair-dyeing event caused a having-red-hair eventuality.

In English, while epistemic *must* is felicitous in cases of abductive inference, WOLL results in infelicity (Winans, 2016). When *-kh* inflects on a high auxiliary (65), it patterns like English epistemic *must*. Contexts are given in (65) based on the examples of abduction discussed in Winans (2016). In (65a), an adapted version of the barbecuing context from Winans (2016), the inferred cause is an ongoing eventuality.[18] In (65b), the inferred cause is a plan for a future eventuality, and the eventuality that triggers the inference is Oyuun's lawnmower being left out. If seeing the lawnmower makes Tengis conclude that Oyuun is going to mow her lawn later, this serves as an explanation for why her lawnmower is outside in the first place. Because this inference is from an eventuality to its cause, this is an instance of abductive reasoning.

(65) a. Context: Tengis knows that his neighbor, Oyuun, grills meats often. Tengis is taking a walk with his friend around the neighborhood, and he smells smoke.

Oyuun odoo makh shar-aj bai-kh bai-**kh**.
Oyuun now meat grill-CVB AUX-*kh* AUX-*kh*
'Oyuun must be grilling meat now.' (Epistemic)

[17] Unlike other statives, copular uses of *baikh* 'to be' can only be interpreted as present with *-nA*. I leave an analysis of why the future-oriented reading is exceptionally unavailable here to future work.

[18] The sentence in (65a) is infelicitous with *-nA*, providing additional support against *-nA* as a non-past tense: while the anti-abductive requirement of (65a) can be accounted for with a WOLL-like modal, it's unclear what causes infelicity if futurity is solely contributed by a non-past.

b. Context: Tengis knows that his neighbor, Oyuun, mows her lawn often. Tengis is taking a walk with his friend around the neighborhood, and he sees a lawnmower by Oyuun's door.

Oyuun udakhgüi züleg-ee khad-akh bai-**kh**.
Oyuun soon grass-REFL.POSS mow-*kh* AUX-*kh*
'It must be the case that Oyuun will mow the lawn soon.' (Epistemic)

In cases of abductive reasoning, WOLL is infelicitous, unlike *must* (Winans, 2016). Along with the prior data, the felicity of *-kh* with abductive inferences indicates that it behaves like an epistemic modal when it is syntactically high.

Based on these six pieces of evidence, the distribution of *-kh* parallels other modals cross-linguistically. I conclude that *-kh* is a modal element with an underspecified flavor. I take the overt realization of *-kh* under negation as evidence for the presence of a covert modal in future-oriented sentences with *-nA*.

5.2.1 *-kh* itself does not encode futurity

One might expect from the modality of *-kh* that future-shifting is also part of its semantics. Under this perspective, the modal and temporal components necessary for future shifting would be collapsed into a single morpheme. However, as seen in (60, 63b, 64a, 65a), *-kh* is compatible with strictly present-oriented contexts. This indicates that *-kh* encodes modality, but not futurity.

The lack of obligatory future shifting with *-kh* is further supported by its behavior in backward counterfactuals, where future-shifting is also not required. In (66), both counterfactuals have backtracked interpretations, made explicit by the inclusion of a present-oriented temporal adverbial in their antecedents and a past-oriented adverbial in their consequents. If *-kh* itself were specified for a future RT and contributed temporal shifting forwards, it would be predicted to mismatch with the adverbials' temporal reference and be infelicitous.

(66) a. Khervee önöödör gazar möstei bai-saŋ bol öchigdör khuitei bai-**kh** bai-saŋ.
if today ground icy AUX-PST COND yesterday cold AUX-*kh* AUX-PST
'If there had been ice on the ground today, it would have been cold yesterday.'

b. Khervee önöödör gadaa chiigtei bai-saŋ bol öchigdör boroo or-**okh** bai-saŋ.
if today outside wet AUX-PST COND yesterday rain enter-*kh* AUX-PST
'If it had been wet outside today, it would have rained yesterday.'

The acceptability of (66) again suggests that *-kh* does not enforce future-shifting, supporting an analysis where *-kh* is a modal lacking temporal specification. However, a fully-developed account of the semantics of conditionals will be necessary to draw more robust conclusions regarding the temporal contribution of *-kh* in backward counterfactuals.

6 Extending the analysis: Negation of the future

I now extend the analysis of affirmative sentences with *-nA* to their negated counterparts. Future readings under negation are discussed in Section 6.1, followed by present readings of statives in Section 6.2. To account for modal-temporal interactions with negation, I give a revised definition of MOD using a selection function semantics (Cariani & Santorio, 2018) .

6.1 Future readings under negation

I begin with future readings of eventives under negation. This is shown in (67), repeated from (53). The present tense *-nA* is unpronounced, and the verbal root is instead inflected with the modal *-kh*.

(67) Margad (margaash) zakhia bich-**ikh-güi**.
Margad (tomorrow) letter write-*kh*-NEG
'Margad won't write a letter (tomorrow).'

A structure for the negated sentence in (67) is provided in (68).

(68) [$_{TP}$ [$_{NegP}$ [$_{ModP}$ [$_{AspP}$ [$_{AspP}$ [$_{vP}$ Margad write letter] $_{Asp}$ PRFV] $_{Asp}$ PROSP] $_{Mod}$ MOD] $_{Neg}$ NEG] $_{T}$ PRS]

In (68), negation is syntactically higher than the modal. However, semantically, negation takes scope above event quantification instead, consistent with other cases of sentential negation cross-linguistically (Acquaviva, 1997; Zeijlstra, 2022). To demonstrate, consider the predicted truth conditions if negation were interpreted in its surface position: the expected meaning for (67) would be that not every world includes a future time containing the duration of a writing-a-letter event by Margad. Put differently, (67) would incorrectly mean that Margad will write a letter at a future time in some worlds, but not all.

This scopal behavior reflects the behavior of WOLL in English. In examples like (69), regardless of whether negation is syntactically above or below WOLL, its truth conditions are unchanged (Thomason, 1970; Copley, 2002; MacFarlane, 2014; Cariani & Santorio, 2018).[19]

(69) a. The symphony will not perform tomorrow.
b. It is not the case that the symphony will perform tomorrow.

Therefore, it is unsurprising that the WOLL-like operator *-kh* is impervious to the scope of negation in Khalkha Mongolian. However, the preliminary version of the semantics for MOD in (37) does not explicitly derive this scopeless behavior, as nothing prevents negation from semantically scoping above it. The scopelessness of English WOLL has been analyzed as the result of a homogeneity presupposition (Copley, 2002), or, alternatively, via a semantics that utilizes a selection function rather than possible world quantification (Cariani & Santorio, 2018). In this paper, I adopt a selection semantics approach to account for these facts.

Unlike an analysis where possible world quantification is restricted via a modal base (see 37), a selection semantics account treats WOLL-like operators as selecting a single world from the modal base via a contextually-supplied selection function σ (Cariani & Santorio, 2018). The constraints in (70) are imposed on σ, necessitating that (i) the selected world verifies the prejacent proposition p and (ii) in cases where the input world w verifies p, w is selected.

[19] WOLL is also scopeless with predicates where negation is lexicalized (Cariani & Santorio, 2018), shown by the truth-conditional equivalence of (i–ii).

(i) Salieri believes that Mozart will fail the test.

(ii) Salieri doubts that Mozart will pass the test.

(70) σ is a selection function iff:
(i) if p is non-empty, $\sigma(w, p) \in p$, and
(ii) if $w \in p$, $\sigma(w, p) = w$.

A revised semantics for MOD is given in (71) for the relevant future-shifting cases. Similar to the preliminary semantics from (37), (71) still shifts the world of evaluation. The primary difference, however, is that this shifting is instead achieved by a selection function, choosing a single world instead of quantifying over worlds. Like before, the modal itself does not contribute future temporal reference by itself, still necessitating a covert PROSP (see 38).

(71) **Final version:**
$[\![\text{-}kh_{\text{CIRC}}]\!]^{\sigma,g,c} = [\![\text{MOD}_{\text{CIRC}}]\!]^{\sigma,g,c} = \lambda p.\lambda t.\lambda w.\ p(\sigma(w, \text{MB}_{\text{CIRC}}(w,t)))$

With this revised modal semantics, truth conditions for the negated eventive example are provided in (72). Here, the interpretation of negation scoping beneath the modal is equivalent to when it scopes above. If negation scopes beneath the modal from (71), its truth conditions are verified when there is no instantiation of the relevant event. This would mean that at $\sigma(w_c, MB_{\text{CIRC}}(w_c,t))$, there is no event of Margad writing a letter at *t"* (see 72). On the other hand, if negation scopes above the modal, its truth conditions are verified when the prejacent proposition does not hold at $\sigma(w, MB_{\text{CIRC}}(w,t))$. For this sentence, this would require that *Margad writes a letter at t"* is false at $\sigma(w_c, MB_{\text{CIRC}}(w_c,t))$. Because the scenarios that satisfy both sets of truth conditions are equivalent, WOLL-like operators appear scopeless with respect to negation.

(72) $[\![(68)_i]\!]^{\sigma,g,c}$ is true iff $\exists$t' [t' = t_c & $\exists$t"[t' < t" & t" $\subseteq$ g(i) & $\neg\exists$e [τ(e) $\subseteq$ t" & write(e, Margad, letter, t", $\sigma(w_c, \text{MB}_{\text{CIRC}}(w_c,t'))$)]]]

The only difference between the truth conditions in (72) and their affirmative counterpart in (42) is the presence of negation. Thus, this analysis captures the interpretive relationship between affirmative and negative constructions in Khalkha Mongolian, despite their morphological asymmetry on the surface.

Moving onto statives, consider the negated sentence shown in (73), repeated from (54b). As previously discussed, this sentence is ambiguous between a future-oriented and present-oriented reading. I will first discuss the future reading, returning to the present reading in Section 6.2.

(73) Erdenechimeg shaltgaan-iig med-**ekh**-**güi**.
Erdenechimeg reason-ACC know-*kh*-NEG
'Erdenechimeg doesn't/won't know the reason.'

A structure for the future-oriented reading of the sentence in (73) is shown in (74).

(74) [$_{\text{TP}}$ [$_{\text{NegP}}$ [$_{\text{ModP}}$ [$_{\text{AspP}}$ [$_{\text{AspP}}$ [$_{\text{vP}}$ Erdenechimeg know reason] $_{\text{Asp}}$ PRFV] $_{\text{Asp}}$ PROSP] $_{\text{Mod}}$ MOD] $_{\text{Neg}}$ NEG] $_{\text{T}}$ PRS]

The truth conditions for the negated, future-shifted structure is given in (75), again utilizing the selection semantics from (71).

(75) $[\![(74)_i]\!]^{\sigma,g,c}$ is true iff $\exists$t' [t' = t_c & $\exists$t"[t' < t" & t" $\subseteq$ g(i) & $\neg\exists$s [τ(s) $\subseteq$ t" & know(s, Erdenechimeg, reason, t", $\sigma(w_c, \text{MB}_{\text{CIRC}}(w_c,t'))$)]]]

Consistent with eventives under negation, the only difference between the affirmative truth conditions in (49) and their negated counterparts is the presence of a negative operator. Thus, the current analysis captures the interpretive relationship between affirmative and negative constructions for the statives, as with the eventives.

6.2 Present readings under negation

Let's return to the observation that negated sentences with statives, like (73), can also receive present-oriented readings. Notably, *-kh* surfaces under negation for both present and future interpretations. A present-oriented reading of *-kh* in this environment appears unexpected if it is modal, as argued above, seemingly rendering the modality of *-kh* vacuous.

However, taking cross-linguistic interactions between negation and verbal morphology into account, this pattern becomes less surprising. Beyond Khalkha Mongolian, other languages have been shown to realize irrealis/future morphology under negation, including in present-oriented sentences (Miestamo, 2005; von Prince et al., 2022; Phillips, 2025). (76) shows such a paradigm from Djambarrpuyŋu, which possesses inflectional categories that encode an (ir)realis distinction (Phillips, 2025).[20] While realis inflection is available for the present affirmative sentence in (76a), its negated counterpart in (76b) requires irrealis inflection. For future sentences (76c–76d), irrealis marking appears regardless of polarity.

(76) **Djambarrpuyŋu** (from Phillips, 2025)

a. ŋarra ga nhä-ma mukulnha
1.SG IMPF.REAL see-REAL aunt.ACC
'I see my aunt (right now).'

b. bäyŋu ŋarra gi nhä-ŋu mukulnha dhiyaŋ bala
NEG 1.SG IMPF.IRR see-IRR aunt.ACC now
'I don't see my aunt right now.'

c. godarr ŋarra dhu nhä-ŋu mukulnha
tomorrow 1.SG FUT see-IRR aunt.ACC
'I'll see my aunt tomorrow.'

d. bäyŋu ŋarra dhu nhä-ŋu mukulnha
NEG 1.SG FUT see-IRR aunt.ACC
'I won't see my aunt tomorrow.'

Why does modal morphology in Djambarrpuyŋu appear in present-oriented sentences under negation? Phillips (2025) accounts for this distribution by proposing that the modal contribution of the irrealis is what leads to its compatibility with nonveridical operators like negation, presupposing that its prejacent is not a metaphysical necessity. One way to extend this to the case of *-kh* is to say that in Khalkha Mongolian, the nonveridical environment induced by negation likewise results in the modal *-kh* surfacing. Under this reasoning, a modal characterization of *-kh* can be maintained in conjunction with its occurrence in present-oriented, negated sentences.

[20] Along with (ir)reality, the Djambarrpuyŋu paradigm contrasts in '(pre)contemporaneity'. In the cyclic tense system of Djambarrpuyŋu, a value of CONTEMPORARY corresponds to both the present and yesterday's past (Phillips, 2025). For our purposes, what matters is that CONTEMPORARY allows present reference.

Even if we set aside the question of why modal morphology appears under negation, the availability of present reference in the first place follows directly from the current analysis due to the selection semantics given for *-kh*. Recall the second constraint that is imposed on the selection function in (70): whenever the input world *w* verifies *p*, *w* itself is selected. As noted by Cariani & Santorio (2018), a consequence of this constraint in unembedded contexts is that the selection function simply returns the world of evaluation, which, in most cases, is the actual world by default. This renders the modal contribution of WOLL semantically vacuous, while still retaining its status as a modal by virtue of its modal base.

I use this property of the selection function to derive the present readings of statives under negation. This rests on a key difference between the proposal for WOLL given by Cariani & Santorio (2018) and the current analysis concerning the source of temporal shifting. While their proposal assumes that WOLL itself introduces quantification over future times, I have argued that temporal shifting stems from a separate component, PROSP, which is divorced from *-kh*. An outcome is that *-kh* can occur in the absence of PROSP, resulting in present readings (see Section 5.2.1).[21] I propose that the present reading of (73) is another such case. Its structure is in (77a), along with its truth conditions in (77b).

(77) a. $[_{TP}\ [_{NegP}\ [_{ModP}\ [_{AspP}\ [_{vP}$ Erdenechimeg know reason$]_{Asp}$ PRFV$]_{Mod}$ MOD$]_{Neg}$ NEG$]_{T}$ PRS$]$

b. $[\![(74)_i]\!]^{\sigma,g,c}$ is true iff $\exists t'$ [$t' = t_c$ & $\neg\exists s$ [$\tau(s) \subseteq t'$ & know(s, Erdenechimeg, reason, t', $\sigma(w_c, MB_{CIRC}(w_c,t'))$)]]

Without PROSP, futurity is not expressed by (77b). Instead, it is entailed that a truth-knowing state is contained within a time that's equivalent to the UT. Since the selection function outputs the world of evaluation per the constraints in (70), a present-oriented reading obtains.

Overall, I have analyzed future readings of sentences with the present tense marker *-nA* as stemming from a covert modal and covert prospective aspect, in lieu of a non-past tense. By positing these covert morphemes, I accounted for the interpretive relationship between affirmative sentences with *-nA* and their negative counterparts with *-kh*. Interactions between negation and modality are modeled through a selection semantics, predicting both the scopelessness of the future and the availability of present readings for statives under negation.

7 Conclusion and implications

In this paper, I have developed an account of future temporal reference in Khalkha Mongolian sentences without overt future marking. I derive futurity not from the contribution of a non-past tense, but instead from the presence of a covert prospective aspect and a covert modal, the latter of which surfaces only under negation. This proposal supports the status of future temporal reference as modal, rather than non-modal: even though the modal operator does not overtly surface alongside VPs marked with *-nA*, its presence can nonetheless be detected. This analysis adds to cross-linguistic work on future reference by contributing a previously unattested pattern of future marking in which the prospective aspect is always covert and the

[21] The modal *-kh*, as well as its covert counterpart (see Section 4), is restricted to environments which invoke irreality, such as negation and the future (von Prince et al., 2022). Thus, no modal appears in the affirmative, present-oriented sentences with statives.

modal, though typically covert, can overtly appear in specific environments (Matthewson et al., 2022; Mucha, 2016; Pancheva & Zubizarreta, 2023; Tonhauser, 2011; Rivero, 2023).

An open issue concerns whether this proposal extends to other attested cases of non-past tenses: are there true instances of non-modal, non-pasts? I have shown that at least in Khalkha Mongolian, an account of *-nA* as a present tense is necessary. Whether or not non-past tenses exist elsewhere, however, is still an empirical question.

What would constitute sufficient evidence to establish that a given tense marker is, in fact, non-past? The availability of both present and future readings on its own is not enough. I conclude by identifying three desiderata for a satisfactory non-past tense account. Recall that a genuine non-past tense is predicted to introduce an RT that is either equivalent to or follows the UT (see Section 3.2.1). Under this characterization, a non-past tense should temporally mirror a non-future tense and be capable of introducing an RT which spans across both the UT and the future. For this reason, if a tense marker is a true non-past, it should parallel the behavior of non-futures in sentences with conjoined subjects (Matthewson, 2006), unlike Khalkha Mongolian *-nA* (see 26b).

In addition, a genuine non-past tense would predict present and future reference to be available across aspects, rather than limiting future readings to the perfective (see Section 3.1.3). Considering the parallel between non-past and non-future once more, both past and present reference is available with the imperfective in the St'át'imcets example from (25). Yet, in Khalkha Mongolian, futurity with *-nA* is disallowed for the imperfective (see 14), where it should otherwise be possible with a true non-past.

A final point concerns the relationship between modality and putative non-past tenses. As shown in Section 3.1.4, sentences with *-nA* license modal subordination, which I have argued is possible via the presence of a covert modal operator. However, if there exists a non-past tense which achieves future reference through purely temporal means (i.e., introducing a non-past RT in the absence of a modal), licensing modal subordination would be unexpected. To claim that a non-past tense is strictly non-modal, it would be necessary to show that modal subordination is disallowed.

Until a tense marker can meet these desiderata, the status of non-past as a category is dubious.

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